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Critical Metrology

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Abstract

Critical quantum metrology leverages the extreme susceptibility of systems near Quantum Phase Transitions (QPTs) to surpass classical precision limits in parameter estimation. This work investigates the Quantum Rabi Model (QRM) as a candidate for such enhanced metrology, establishing that finite-component systems can mimic thermodynamic criticality within specific regimes, as the classical oscillator limit for the QRM. Theoretical analysis employing the Quantum Fisher Information (QFI) and Cramér-Rao Bound (CRB) indicates that estimation error theoretically vanishes at the critical point. To characterize this behavior, effective Hamiltonians are derived via first- and second-order Schrieffer-Wolff (SW) transformations. Second-order corrections refine the analytical model without altering the fundamental nature of the transition. Crucially, both approaches successfully demonstrate the divergence of the QFI in the normal phase.

However, due to the inversion of the effective potential and macroscopic photon occupation in the superradiant phase, standard analytical and numerical methods break down. To address this, we introduce a numerical strategy utilizing a unitary displacement transformation. By shifting the Fock basis by a variational parameter corresponding to the superradiant mean field, the method restores numerical convergence and allows for accurate high-fidelity diagonalization. The resulting simulations confirm the divergence of the QFI with a critical exponent of -2 and identify a tendency change in the entanglement entropy. These findings substantiate the potential of the finite-size QRM for critical sensing applications, bridging the gap between finite-system dynamics and critical many-body phenomena.

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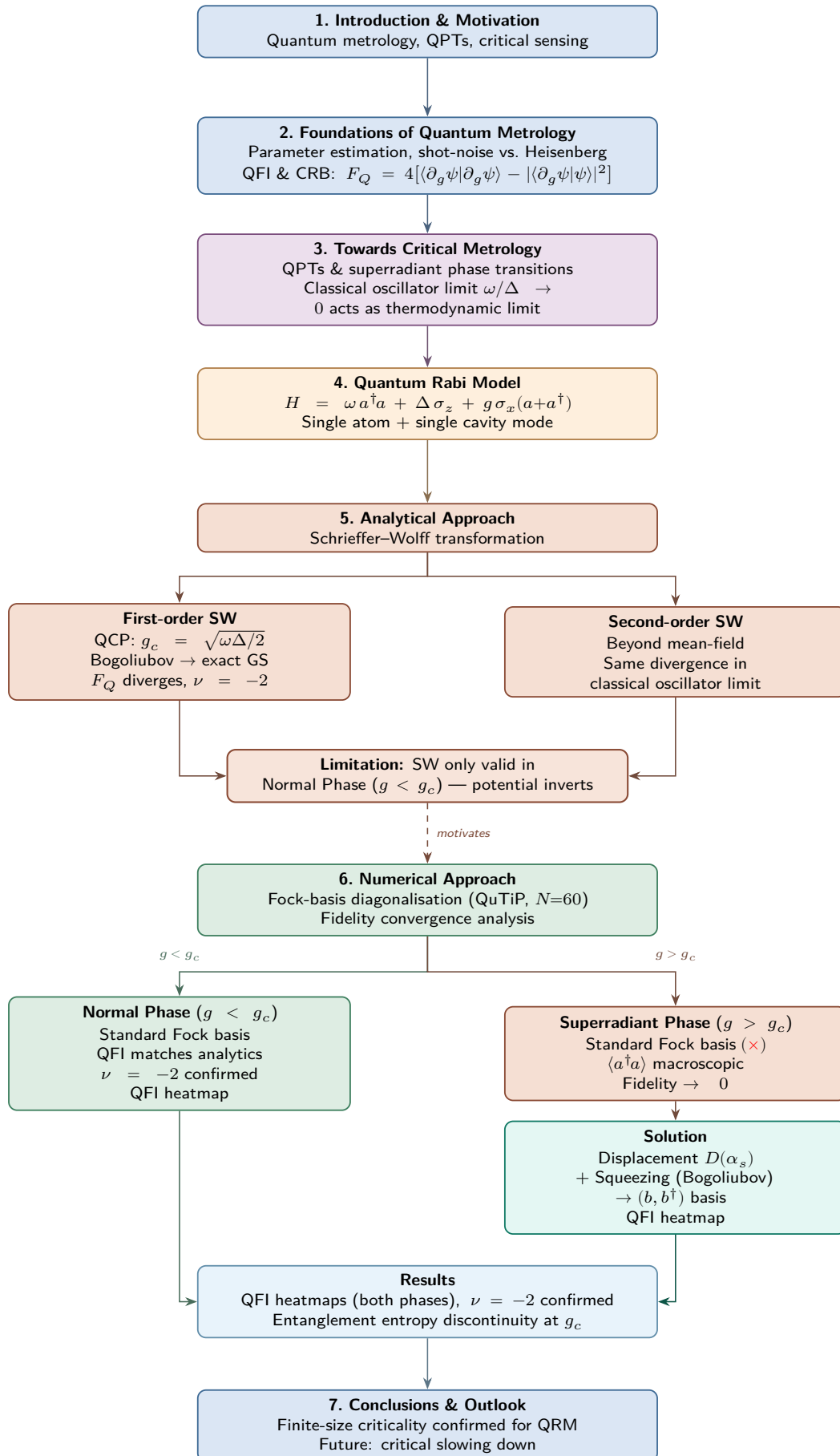
Special thanks to my colleagues Erik and Jaime for walking me through the correct usage of GitHub, keeping my work structured since day one.

Finally, even if I am falling into a cliché, I would like to thank my family and friends for always being there.

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1 Conceptual flow-chart



2 Introduction

The discipline of metrology constitutes a basis for scientific inquiry and technological advancement. Metrological activities have traditionally been concerned with classical or semi-classical systems, i.e., mechanical or optical systems whose behavior is determined by classical physics laws. With the current second Quantum Revolution, we are now in an era where it is possible to precisely observe and skillfully manipulate the behavior of many-body quantum systems. This expertise has driven the development of quantum metrology, a discipline committed to leveraging native quantum attributes to transcend the precision boundaries that classical physics prescribes. Far from being an abstract activity, quantum metrology provides an essential resource for a wide variety of applications, from fundamental investigations of physical laws to the implementation of exquisitely sensitive sensors in areas such as magnetometry and interferometry [1]. Using quantum effects to enhance measurement is not just a step forward, it is a complete rethink of how we define precision. Instead of relying solely on classical tools and techniques, quantum metrology taps into the fundamental properties of quantum systems, offering the potential for breakthroughs across science and technology.

One of the most promising strategies to unlock this enhanced precision relies on exploiting systems near instability [2], specifically at Quantum Phase Transitions (QPTs). Unlike classical phase transitions driven by thermal fluctuations, QPTs occur at absolute zero and are governed entirely by quantum fluctuations that abruptly alter the system's Ground State (GS). As the spectral gap vanishes, the ground state undergoes a macroscopic reorganization [3], rendering the system exceptionally susceptible to infinitesimal changes in the driving parameters. This very feature makes QPTs an invaluable metrological resource. However, a major theoretical inconvenience arises: general QPTs exhibit true critical behavior only in the thermodynamic limit, where the system size approaches infinity, severely restricting the available numerical methods.

One prominent class of QPTs is Superradiant Phase Transitions (SPTs) [4], where a system shifts from a vacuum normal phase to a superradiant phase with macroscopic photon occupation. For decades, it was believed that the sharp non-analyticities of SPTs—and QPTs in general—strictly required the thermodynamic limit, thereby restricting studies to infinite-size models such as the Dicke model. However, a major realization in recent years is that strict criticality can also emerge in finite-sized systems, such as the Quantum Rabi Model (QRM) [5]. The QRM involves just a single two-level atom interacting with a single cavity mode. Here, the critical behavior is recovered in the so-called classical oscillator limit, where the atomic transition frequency becomes much larger than the cavity mode frequency. This frequency ratio acts as an effective thermodynamic limit, demonstrating that criticality can also be achieved in finite-component systems through an appropriate parameter scaling.

Having established the critical behavior of the Quantum Rabi Model, a natural question arises: how can we leverage its phase transition to enhance estimation precision? The core idea of critical metrology is to exploit the system's extreme sensitivity near the critical point as a measurement resource, transforming this heightened responsiveness into a powerful metrological tool. However, there is a catch: as the system nears the critical point and the energy gap shrinks, it takes longer and longer to reach its GS. This is known as critical slowing down [2]. While critical slowing down is an inevitable dynamical consequence, this thesis focuses exclusively on the static equilibrium properties of the

QRM.

Following the framework that Garbe et al. proposed [2], we leverage the quantum phase transition of the QRM to study parameter estimation near the critical point. More specifically, we employ the Quantum Fisher information (QFI) and the Cramér-Rao bound (CRB), pillars of quantum metrology. Besides performing analytical calculations, we conduct numerical simulations to cross-check and complement the theoretical work.

The code used for the numerical analysis and the generation of several images is available in a public GitHub repository, see Ref. [6]. There is substantially more work in that repository than appears in this text, due to the need for synthesis. Most of the numerical simulations were performed using QuTiP [7].

Throughout this thesis, unless otherwise explicitly stated, we adopt natural units where the reduced Planck constant is set to unity ($\hbar = 1$). We also make use of many acronyms for readability¹.

3 Foundations of Quantum Metrology

In the heart of classical metrology lies the idea of parameter estimation, more precisely, of maximizing the precision of the estimation. Suppose we would like to estimate a generic parameter θ ; then, the smallest possible error scales as $(\Delta\theta)^2 \sim 1/(\nu N)$ for a classical experiment [1], where ν represents the number of times the experiment is repeated, and N denotes the sample size. This relation is called **shot-noise scaling**.

However, when quantum mechanics comes into play, this framework can be fundamentally bypassed. In the quantum regime, the interpretation of N becomes somewhat obscure and must be reformulated; nevertheless, we can adapt N to represent the effective number of physical resources involved in the sample. Under these conditions, the **shot-noise scaling** can be surpassed, reaching what is called **Heisenberg-scaling**, $(\Delta\theta)^2 \sim 1/(\nu N^2)$ [1]. This means that quantumness is needed for greater precision. Notice that minimizing the error is equivalent to enhancing precision; we will use both phrases interchangeably. Furthermore, even greater scaling is feasible under certain conditions, but we will not delve into that (see e.g., [8, 9]).

We will now understand how to proceed with a simple metrological task before jumping into the more advanced techniques we will employ throughout the work: the QFI and the CRB. It is worth emphasizing that this work focuses exclusively on fundamental theoretical precision limits. Saturating these ultimate bounds experimentally requires applying inference techniques asymptotically, such as Maximum Likelihood Estimation (MLE), to the actual measurement data [10].

¹**Some useful acronyms:** Probability Distribution Function (PDF), Error Propagation Formula (EPF), Symmetric Logarithmic Derivative (SLD), Quantum Phase Transitions (QPTs), Superradiant Phase Transitions (SPTs), Quantum Fisher Information (QFI), Cramér-Rao Bound (CRB), Quantum Rabi model (QRM), Quantum Critical Point (QCP), Rotating Wave Approximation (RWA), Schrieffer-Wolff (SW), Ground State (GS), Positive operator valued-measure (POVM), Projection valued measure (PVM), Baker-Campbell-Hausdorff (BCH).

3.1 Observable-dependent bounds on estimation precision

To understand the necessity of more advanced theoretical tools, it is illustrative to first examine the standard approach to parameter estimation. Consider a fundamental metrological task: estimating an unknown parameter θ (such as the strength of a magnetic field). For an ensemble of N particles, the system's time evolution is typically governed by a unitary operator $U_\theta = e^{-iH\theta}$, where H is the relevant Hamiltonian driving the dynamics.

In this standard approach, information about θ (encoded in H) is extracted by measuring a specific Hermitian observable, which we will call M . According to the standard Error Propagation Formula (EPF), the precision of our estimation is bounded by [1]:

$$(\Delta\theta)^2 = \frac{(\Delta M)^2}{|\partial_\theta \langle M \rangle|^2}, \quad (1)$$

where $\langle M \rangle$ is the expectation value of the selected observable and $(\Delta M)^2$ is its uncertainty or variance.

If the system is prepared in a standard and uncorrelated way—such as a fully polarized ensemble of spins—the variance scales in a way that bounds the precision by the classical shot-noise limit [1]. However, quantum mechanics allows us to bypass this classical restriction. By preparing the system in highly entangled configurations [11, 12], we can selectively reduce the variance of the measured observable. These intrinsic quantum correlations enable the precision to surpass classical bounds, potentially achieving Heisenberg scaling [13].

While calculating the precision via Eq. (1) is intuitive, it presents a fundamental theoretical limitation: it heavily relies on our ability to guess or derive the optimal observable M . This leads us to a crucial question: how can we determine the absolute maximum precision achievable using a quantum state, regardless of the specific measurement chosen? To answer this, we must avoid individual observables and focus on the information geometry of the state itself, leading us to the QFI and the CRB. Relying on these tools becomes essential in complex quantum systems, particularly those approaching a phase transition, where identifying an optimal measurement scheme is highly non-trivial.

3.2 QFI and the CRB

In statistical inference, the estimation of a parameter θ is performed using an estimator $\hat{\theta}$, which is fundamentally a function of the observed data [14]. A standard measure of the estimator's performance is the Mean Squared Error (MSE), defined as the expected value of the squared difference between the estimator and the true parameter value. The MSE can be analytically decomposed into two components: the variance of the estimator and the square of its bias [14],

$$\text{MSE}(\hat{\theta}) = \mathbb{E}_\theta [(\hat{\theta} - \theta)^2] = \text{Var}_\theta(\hat{\theta}) + (\text{Bias}_\theta(\hat{\theta}))^2. \quad (2)$$

For an **unbiased** estimator ($\text{Bias}_\theta(\hat{\theta}) = 0$), the MSE reduces to the variance [14]. Consequently, minimizing the variance directly minimizes the overall estimation error.

It is worth noting a mathematical nuance: restricting our search strictly to unbiased estimators does not universally guarantee the absolute lowest possible MSE. A biased estimator with a significantly reduced variance could, in principle, yield a smaller combined error and thus outperform an unbiased one [14]. To find the absolute lower bound on this variance (and thus the minimum error for unbiased estimators) regardless of the specific measurement chosen, we must delve into the QFI and the CRB.

3.2.1 Classical Fisher information

To understand the QFI, we first need to introduce its classical counterpart. Let us consider the problem of estimating a parameter θ by measuring a quantity M . Let us assume that the relationship between θ and M is given by a Probability Distribution Function (PDF) $f(x; \theta)$, which for a fixed θ gives the probability distribution of the x values of M [1, 15].

To get an estimation of θ given x , we construct an unbiased estimator $\hat{\theta}(x)$. Under these conditions, the precision of all the possible estimators exhibits a lower bound, the CRB [1]. The following inequality must be true for the unbiased estimator

$$\text{Var}_\theta(\hat{\theta}) := \mathbb{E}_\theta((\hat{\theta} - \theta)^2) \geq \frac{1}{\nu F(\theta)}, \quad (3)$$

where we have assumed for the sake of simplicity that θ is a single parameter. Here, ν represents the number of independent measurements that we have conducted. From now on, we will take $\nu = 1$ for simplicity [16].

The factor $F(\theta)$ from Eq. (3) is the well known Fisher information defined as [1, 10]

$$F(\theta) = \int \left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right)^2 f(x; \theta) dx = \int \frac{1}{f(x; \theta)} \left(\frac{\partial f(x; \theta)}{\partial \theta} \right)^2 dx. \quad (4)$$

The Fisher information quantifies how much the probability distribution function changes in response to variations in the parameter θ . In essence, it measures how much information the observed data provides about that parameter [17]. As a quick side note, it is interesting to see that we can obtain the error propagation formula from this definition (see the full derivation and other details in the supplementary material [6]).

3.2.2 Quantum Fisher information (QFI)

Let us go back to classical Fisher information, where we wished to estimate θ through the measurement of M . It is imperative to emphasize that we do not know which the optimal observable M is. However, despite this lack of knowledge, we can still establish an upper bound for the Fisher information. This is what we will do next.

Information about the target parameter is extracted by probing the quantum state of the system using a Positive Operator-Valued Measure (POVM) defined by a set of operators $\{\Pi_x\}$. Operators satisfying $\sum_x \Pi_x = \mathbb{I}$ and $\Pi_x \geq 0$, where each operator is associated with a specific outcome x [16]. That being the case, the probability of obtaining x , for a given θ and quantum state ϱ_θ is

$$f(x; \theta) = \text{Tr}(\varrho_\theta \Pi_x), \quad (5)$$

which is a consequence of the Born rule. Equation (5) can be used to build an unbiased estimator. Analogously to the classical problem, the phase estimation precision is bounded for the quantum case, in this case by the quantum CRB [1]

$$(\Delta\theta)^2 \geq \frac{1}{F_Q(\varrho_\theta)}, \quad (6)$$

with $F_Q(\varrho_\theta)$ being the QFI. To use that expression, we must have a closed formula for the QFI, which is the supremum of the Fisher information. The computation of the QFI is as follows:

We need to maximize the Fisher information over all the POVMs. Following general literature (e.g., [10]), we define the Symmetric Logarithmic Derivative (SLD) operator L_θ implicitly as

$$\frac{L_\theta \varrho_\theta + \varrho_\theta L_\theta}{2} = \frac{\partial \varrho_\theta}{\partial \theta}. \quad (7)$$

From this follows that

$$\partial_\theta f(x; \theta) = \text{Tr}[\partial_\theta \varrho_\theta \Pi_x] = \text{Re}(\text{Tr}[\varrho_\theta \Pi_x L_\theta]), \quad (8)$$

with which we can write the classical Fisher information from Eq. (4) as

$$F(\varrho_\theta) = \int \frac{\text{Re}(\text{Tr}[\varrho_\theta \Pi_x L_\theta])^2}{\text{Tr}[\varrho_\theta \Pi_x]} dx. \quad (9)$$

It can be proved [10] that the Fisher information for a quantum measurement is bounded:

$$F(\varrho_\theta) \leq \text{Tr}[\varrho_\theta L_\theta^2] = \text{Tr}[\partial_\theta \varrho_\theta L_\theta]. \quad (10)$$

That bound is what we call the QFI,

$$F_Q(\varrho_\theta) = \text{Tr}[\varrho_\theta L_\theta^2]. \quad (11)$$

This is the maximum achievable information from a parameter θ via quantum probes. We set out to find the SLD to obtain a less obscure expression for the QFI. Equation (7) is a Lyapunov equation whose general solution for L_θ is [10, 16]

$$L_\theta = 2 \int_0^\infty e^{-t\varrho_\theta} \partial_\theta \varrho_\theta e^{-t\varrho_\theta} dt. \quad (12)$$

Note that a valid density matrix must be Hermitian, which has been applied in Eq. (12). Now, we assume the spectral decomposition of $\varrho_\theta = \sum_i p_i |\psi_i\rangle\langle\psi_i|$, by which the SLD reads

$$L_\theta = 2 \sum_{i,j} \frac{\langle\psi_j|\partial_\theta \varrho_\theta|\psi_i\rangle}{p_i + p_j} |\psi_j\rangle\langle\psi_i|. \quad (13)$$

For the case of a pure state, Eq. (13) simplifies to [18]

$$L_\theta = 2 (|\psi\rangle\langle\partial_\theta \psi| + |\partial_\theta \psi\rangle\langle\psi|), \quad (14)$$

which, if plugged into the definition of the QFI, see Eq. (11), yields [16, 2]

$$F_Q(\varrho_\theta) = 4 [\langle\partial_\theta \psi|\partial_\theta \psi\rangle - \langle\partial_\theta \psi|\psi\rangle\langle\psi|\partial_\theta \psi\rangle]. \quad (15)$$

Using Eq. (15), we can compute the QFI for any quantum state; it is usual, nonetheless, to employ the family ϱ_θ connected to the GS.

4 Towards Critical Metrology

Up to this point, we have introduced the pillars of quantum metrology. It is timely to answer *why* we will apply these principles to systems close to a phase transition, and *what* it is that we find promising in them.

Throughout this chapter, we will explore QPTs and introduce the phenomena of spatial and time correlation divergence near the critical point. We will then introduce the concept of superradiance and Superradiant Phase Transitions (SPTs), which will come in handy later on.

4.1 Quantum phase transitions (QPTs)

In this subsection, we will define QPTs and classification schemes and examine where the promising enhanced responsiveness comes from.

QPTs occur when the variation of a physical parameter, rather than temperature, drives a system from one phase to another with fundamentally different physical properties. Ideally taking place at absolute zero, these transitions stem from quantum fluctuations in the energy spectrum dictated by the Heisenberg uncertainty principle [19, 3]. Standard QPTs—though there are exceptions—fit well into the Landau theory of phase transitions, where an order parameter discriminates distinct phases. This order parameter is a thermodynamic property that remains non-zero in the ordered phase and vanishes in the disordered phase [20].

To formally classify these transitions, a general criterion was initially put forward by Ehrenfest, who associated the degree (order) of the phase transition with the lowest derivative of the free energy that is discontinuous at the transition point. While the discovery of certain exotic transitions eventually led to a simplified binary classification—first-order and continuous phase transitions—the Ehrenfest framework remains a powerful adapted tool to define the order of a transition when nonanalyticities in the free energy are encountered. First-order transitions involve a discontinuity in the first derivative, whereas continuous transitions present discontinuities in higher-order derivatives. We will work with a continuous QPT, but there are first-order QPTs as well [4].

The defining hallmark of such continuous transitions is the emergence of collective behavior and scale invariance at the critical point. Instead of individual constituents fluctuating independently, the system begins to behave as a highly interconnected whole. This macroscopic reorganization is microscopically manifested through long-range spatial correlations, meaning that a perturbation in one region of the system directly affects increasingly distant parts. The characteristic spatial scale of these correlated fluctuations, known as the correlation length ξ , diverges as the system approaches the critical point according to the power law [21]:

$$\xi \propto |t|^{-\nu}, \quad (16)$$

where ν is the correlation length critical exponent and t is a dimensionless parameter that quantifies how close the system is to the critical point. For a quantum phase transition driven by a non-thermal control parameter g (such as a magnetic field or a coupling constant), this distance is naturally defined as $t = |g - g_c|/g_c$, where g_c is the critical value

at which the transition occurs. These parameters arise from Quantum and statistical Field Theory; we will not get into their derivation, but we emphasize their meaning and consequences.

The crux of the matter is closely related to the divergence displayed by Eq. (16). Measurements in physics, and especially in quantum mechanics, are inherently invasive. In order to make a measurement, we must apply a perturbation to the system. Were we to apply a perturbation to a localized constituent of our system, it would spread to the whole system due to the divergence of the correlation length scale. This means that when the system recovers equilibrium, we would be effectively measuring the response of the entire system, not just of the individual constituent we perturbed. Here is where the aforementioned arbitrarily high susceptibility stems from: by just extending the boundaries of our system, we get increasingly heightened precision.

Apart from spatial correlations, we must also consider time. Here we define the correlation time τ_c . This timescale effectively quantifies how rapidly perturbations decay and the system recovers thermodynamic equilibrium. When we approach the critical point, it diverges as [21]

$$\tau_c \propto \xi^z \propto |t|^{-\nu z}, \quad (17)$$

where z is the dynamical critical exponent. The fact that fluctuations take “infinite” time to die off is burdensome. We have seen that in the vicinity of a phase transition, a very localized perturbation spreads to the entirety of the system, and that we just need to wait for equilibrium to be recovered to make an enhanced measurement. Of course, critical slowing down is inevitable, but we emphasize again that we will not concern ourselves with it, as it is the static enhanced behavior near criticality that we wish to study.

Bear in mind that the previous discussion is fundamentally applicable to systems where the notion of distance is well-defined. It is pertinent to remember now that the QRM is a fully finite-sized system. We will prove very soon, however, that the static critical behavior—specifically the enhanced susceptibility—is perfectly replicated in our case study.

4.1.1 QPTs: Ground state properties

The existence of a quantum phase transition is primarily dependent on the Hamiltonian of the system and its GS. Following Sachdev’s explanation [19], we consider a Hamiltonian of the form $H = H_1 + gH_2$, where g is a dimensionless coupling parameter.

Two fundamental conditions are required for a QPT to occur:

1. The GSs of H_1 and H_2 must be different, creating a competition between distinct phases².
2. The terms must not commute, $[H_1, H_2] \neq 0$. This ensures that quantum fluctuations driven by the uncertainty principle prevent the system from simply staying in a single eigenstate of H_1 or H_2 across the transition. They do not share a basis where both are diagonalizable.

²This model of competing terms is highly intuitive but not always applicable. Furthermore, it only explains the existence of 1 QCP, but there might be more.

In the limit $g \rightarrow \infty$, H_1 is negligible, and the GS is determined by H_2 . Conversely, when $g \rightarrow 0$, the physics is dominated by H_1 . Therefore, there must be a critical value $g = g_c$. At this Quantum Critical Point (QCP) the spectral gap—the energy difference between the ground and first excited state—closes, and the GS energy exhibits a non-analyticity. Physically, the domain has been shifted from one part of the Hamiltonian to the other.

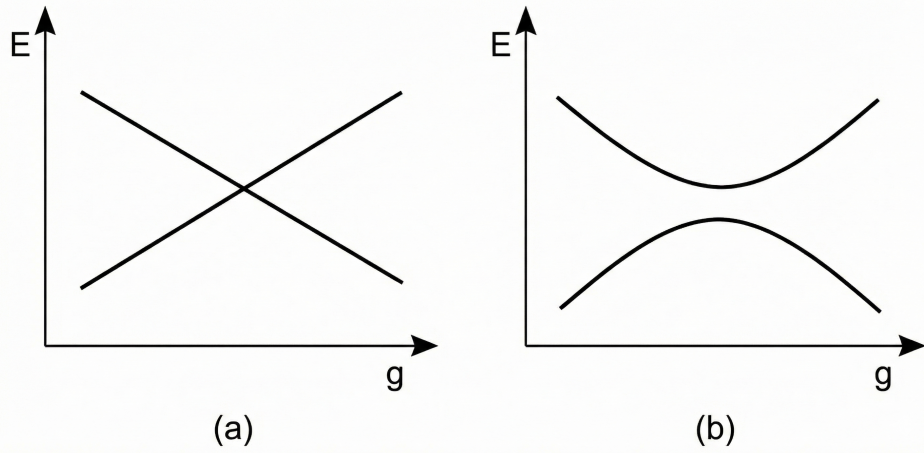


Figure 1: Energies of the GS and excited state as a function of g . (a) A sharp level-crossing. (b) The general case where there is an “avoided level-crossing” due to quantum fluctuations. Image edited from original version at [19].

Strictly speaking, a sharp level crossing—and thus a true non-analytic QPT—occurs only in the thermodynamic limit ($N \rightarrow \infty$). In finite systems, the non-commutativity of the Hamiltonian terms prevents the spectral gap from strictly closing. This leads to an “avoided level crossing”, resulting in a smooth crossover instead of a sharp phase transition, as displayed in Figure 1(b) [19]. Naturally, as the system size increases, this gap minimum sharpens, asymptotically recovering the macroscopic critical behavior. At least, that is what was previously thought.

The interest in finite-sized systems revived when quantum phase transitions with no apparent thermodynamic limit were found in simple quantum optical models [5]. It is now known that a wide class of quantum optical models presents superradiant quantum phase transitions of this form [4]. These cases show that the essence of critical behavior is not strictly tied to the particle number but rather to a scaling of parameters that contributes to the closing of the spectral gap. This breakthrough underpins the entirety of the upcoming work. Let us prove this.

4.2 An example of QPTs: SPTs

First, we should define what superradiance is, as it was originally introduced by Dicke in 1954 [18, 22]. Superradiance was first introduced in the study of the spontaneously emitted photons from a collection of atoms. This phenomenon can be intuitively understood by picturing each atom as a tiny antenna emitting electromagnetic waves. In a typical scenario, thermally excited atoms emit light randomly, and the intensity of the emitted light is directly proportional to the number of atoms, denoted as N . However, a remarkable deviation occurs when these atomic “antennas” radiate coherently and in phase with each other. In such a collective state, the net electromagnetic field generated

is proportional to N , which, in turn, causes the emitted intensity to scale quadratically with the number of atoms, as N^2 . This collective behavior results in the atoms radiating their stored energy N times faster than they would if they were emitting incoherently. Dicke coined this enhanced, anomalous radiance “superradiance” [22].

4.2.1 SPT and superradiant phase

While Dicke’s original formulation described a strictly dynamical phenomenon, the concept has since evolved to encompass static equilibrium properties as well. Light-matter interacting systems can undergo a quantum phase transition where their fundamental nature shifts. Specifically, as the coupling surpasses a critical threshold, the system transitions from a normal phase—characterized by a negligible bosonic occupation such that the relevant order parameter is strictly zero—to a superradiant phase. In this regime, the GS spontaneously acquires a permanent, macroscopic photon population.

Consequently, a superradiant phase is mathematically characterized by a macroscopic average number of photons, meaning that the order parameter $\langle a^\dagger a \rangle / C$, where C indicates the scaling, remains finite in the $C \rightarrow \infty$ limit. Crucially, C does not necessarily have to be the number of constituents in the system [4]. Proving this mathematically is exactly what allows the classical oscillator limit of the finite-sized Rabi model to successfully exhibit a phase transition.

To gain insight into this matter, it is highly instructive to invoke the quantum Dicke model (the interaction of N spins with a single cavity mode) [4]:

$$H = \omega a^\dagger a + \sum_{i=1}^N \frac{g}{\sqrt{N}} \sigma_{ix} (a + a^\dagger) + \sum_{i=1}^N \Delta \sigma_{iz}. \quad (18)$$

As usual, a and $a^\dagger$ denote annihilation and creation operators for a bosonic mode, while the Pauli matrices also carry a spin index. The parameters of this family are ω (photon energy), 2Δ (spin energy), g (coupling strength), and N (any positive integer, representing the number of spins).

The QRM, which will be formally derived and explored in the next chapter, corresponds precisely to the extreme case of $N = 1$. However, before diving into the physical derivation of the single-atom model, we will use the generalized Dicke model to understand how criticality can emerge without an infinite number of particles. We will follow the treatment from [4].

Instead of limiting our analysis to the $\langle a^\dagger a \rangle / N$ order parameter case (where $N \rightarrow \infty$ would be the regular thermodynamic limit), we rewrite the Hamiltonian above in terms of the generic scaling parameter C . This way, the order parameter becomes $\langle a^\dagger a \rangle / C$. The Hamiltonian takes on the following kind,

$$H = \Delta \sum_{i=1}^N \left[\frac{\omega C}{N \Delta} \frac{a^\dagger a}{C} + \frac{g \sqrt{C}}{\sqrt{N} \Delta} \sigma_{ix} \frac{(a + a^\dagger)}{\sqrt{C}} + \sigma_{iz} \right]. \quad (19)$$

We recall that for a phase transition to arise, the Hamiltonian must be comprised of competing terms. Therefore, as we scale the system parameters, we must simultaneously adjust the interaction strength so that neither term vanishes or diverges when

$C \rightarrow \infty$. Mathematically, the effective couplings $\Omega = \omega C/N\Delta$ and $\gamma = g\sqrt{C}/\Delta\sqrt{N}$ need to remain finite. Consequently, the region of parameter space of interest is given by $N\Delta/\omega \rightarrow \infty$ while $g^2/\Delta\omega$ is kept finite.

One way to achieve this regime is by letting $N \rightarrow \infty$ while keeping all other parameters fixed, which corresponds to the standard thermodynamic limit. The alternative option, coined the **classical oscillator limit**, is obtained by taking $\Delta/\omega \rightarrow \infty$ and $g/\omega \rightarrow \infty$, while γ and Ω remain finite [4].

For a finite-sized system such as the QRM (where $N = 1$), the order parameter scales as $\langle a^\dagger a \rangle / (\Delta/\omega)$. By mathematically establishing that the classical oscillator limit ($\Delta/\omega \rightarrow \infty$) plays the exact role of a thermodynamic limit, we have proven that the QRM is capable of exhibiting a rigorous SPT. To find the exact QCP, we will eventually use a mean-field approach to see where a major shift occurs in the properties of the GS. However, with this fundamental scaling argument secured, we are now perfectly equipped to proceed to the next chapter to uncover the physical origins of the QRM itself.

5 Uncovering the Rabi Model

5.1 The Semiclassical Rabi Model

The study of light-matter interaction finds one of its most fundamental pillars in the work of Isidor Isaac Rabi. In the late 1930s, Rabi developed a method to measure the magnetic moments of atomic nuclei using molecular beams, a technique for which he was awarded the Nobel Prize in Physics in 1944 [23]. His insight was to use an oscillating electromagnetic field to induce transitions between spin states, a phenomenon now known as "Rabi flopping" or Rabi oscillations. This semiclassical description, treating the atom quantum mechanically and the field classically, laid the theoretical groundwork for Nuclear Magnetic Resonance (NMR) and modern quantum information control [24].

Consider a two-level system with ground state $|g\rangle$ and excited state $|e\rangle$, separated by a transition frequency ω_0 . When driven by a near-resonant classical field of frequency ω , the total Hamiltonian is given by $H = H_0 + H_{\text{int}}(t)$. The interaction term, typically of the dipole form, couples the states with a strength proportional to the field amplitude.

To solve the dynamics, we propose a general state vector,

$$|\psi(t)\rangle = C_g(t)e^{-iE_g t} |g\rangle + C_e(t)e^{-iE_e t} |e\rangle. \quad (20)$$

By substituting this ansatz into the time-dependent Schrödinger equation and applying the Rotating Wave Approximation (RWA)—which allows us to neglect terms oscillating at high frequencies $\omega + \omega_0$ —the dynamics simplify to a set of coupled first-order differential equations [25].

Skipping the algebraic details of the integration, the system admits an exact analytical solution for the population coefficients [25]. If the system is initially in the GS ($C_g(0) = 1$, $C_e(0) = 0$), the probability of finding the atom in the excited state at time t is given by,

$$P_e(t) = |C_e(t)|^2 = \left(\frac{\Omega}{\Omega_R} \right)^2 \sin^2 \left(\frac{\Omega_R t}{2} \right). \quad (21)$$

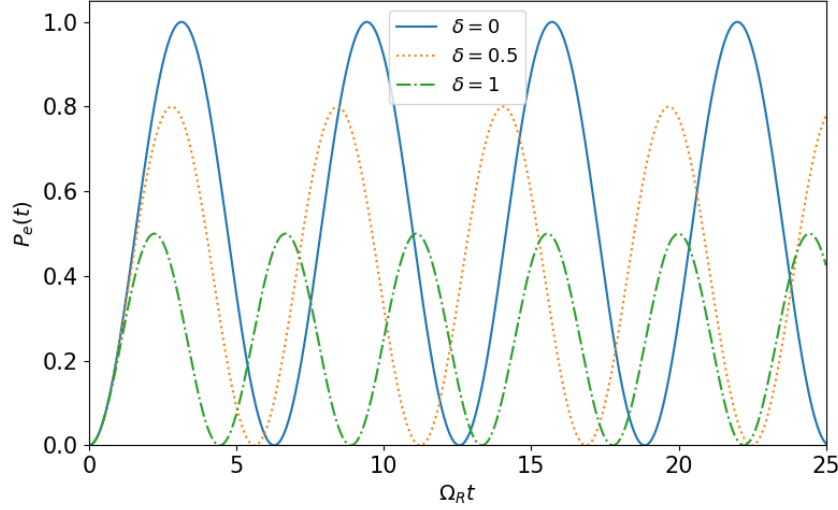


Figure 2: Rabi oscillations of the probability $P_e(t)$ as a function of time for three detuning regimes: resonance ($\delta = 0$), intermediate detuning ($\delta = 0.5$), and large detuning ($\delta = 1$). Note how the generalized Rabi frequency $\Omega_R = \sqrt{\Omega^2 + \delta^2}$ increases with detuning, leading to faster oscillations but reduced amplitude.

Here, Ω represents the coupling strength (often called the bare Rabi frequency), and Ω_R is the generalized Rabi frequency, defined as:

$$\Omega_R = \sqrt{\delta^2 + \Omega^2}, \quad (22)$$

where $\delta = \omega_0 - \omega$ is the detuning between the atomic transition and the external driving field.

This result highlights the critical importance of resonance. As shown in Fig. 2, complete population transfer ($P_e = 1$) is only possible in the resonant limit where $\delta = 0$. In the context of NMR, this precise control of population transfer via pulse duration allows for the manipulation of nuclear spins, forming the basis of pulse sequences used in magnetic resonance imaging (MRI).

5.2 Intuitive derivation of the QRM Hamiltonian

We have already covered the semiclassical model, and consequently, we are ready to take the leap and quantize the field. The QRM describes the interaction of a quantized field with a two-level atom in a cavity and is characterized by three terms [25]

$$H_{\text{Rabi}} = H_{\text{Field}} + H_{\text{Atom}} + H_{\text{Inter}}. \quad (23)$$

The first term corresponds to the field Hamiltonian, the second to the one of the two-level atom, and the last one to the interaction. We will treat them separately.

For the atomic Hamiltonian, we can choose the zero level energy to be exactly halfway between the ground and excited level, so

$$H_{\text{Atom}} = \Delta(|e\rangle\langle e| - |g\rangle\langle g|) = \Delta\sigma_z, \quad (24)$$

where Δ corresponds to two times the qubit frequency, sticking to the same notation used beforehand for the Dicke model.

The field Hamiltonian arises from the quantization of the electromagnetic field. For a one-dimensional cavity of length L along the z direction, the expressions for the EM-field that satisfies the boundary conditions (i.e., vanishing at $z = 0$ and $z = L$) and Maxwell's equations are [25]:

$$\begin{aligned}\mathbf{E}(z, t) &= \sqrt{\frac{2\omega^2}{V\varepsilon_0}} q(t) \sin(kz) \mathbf{e}_x, \\ \mathbf{B}(z, t) &= \frac{\mu_0\varepsilon_0}{k} \sqrt{\frac{2\omega^2}{V\varepsilon_0}} \dot{q}(t) \cos(kz) \mathbf{e}_y.\end{aligned}\tag{25}$$

Here q is a time-dependent factor having the dimension of length, and V is the effective volume of the cavity. It is usual to rewrite $\dot{q}$ as p . A very important remark is that we only take into consideration a single cavity mode, because we assume that the other modes are far away in frequency from the particle's transition of interest.

In this situation, one can compute the classical field energy as

$$H_{\text{Field}} = \frac{1}{2} \int dV \left[\varepsilon_0 \mathbf{E}^2(z, t) + \frac{1}{\mu_0} \mathbf{B}^2(z, t) \right]. \tag{26}$$

From Eq. (26) the field Hamiltonian becomes $H_{\text{Field}} = \frac{1}{2}(p^2 + \omega^2 q^2)$, which corresponds to a unit mass harmonic oscillator. This Hamiltonian can be rewritten in terms of the bosonic creation and annihilation operators as $H_{\text{Field}} = \omega a^\dagger a$, where we have dropped the zero-point energy, as it does not contribute to the system's dynamics.

With the field and atomic Hamiltonians derived, we shift our attention to the most intriguing of them all, the interaction Hamiltonian.

We recover the expressions from Eq. (25). We have $q(t)$ and $p(t)$, which are Hermitian operators. We promote them to operators with the second quantization and relate them to the non-Hermitian bosonic annihilation and creation operators $a = (2\omega)^{-1/2}(\omega q + ip)$ and $a^\dagger = (2\omega)^{-1/2}(\omega q - ip)$. From this, the expression of the electric field turns into

$$\mathbf{E} = \left(\frac{\omega}{\varepsilon_0 V} \right)^{\frac{1}{2}} (a + a^\dagger) \sin(kz) \mathbf{e}_x,$$

where $\mathbf{e}$ denotes an arbitrary unit vector. Now, to find the interaction Hamiltonian, we apply the dipole approximation, that is $\lambda \gg r_{\text{dipole}}$ (dimension of the system). We get

$$H_{\text{Inter}} = -\mathbf{D} \cdot \mathbf{E} = -D \left(\frac{\omega}{\varepsilon_0 V} \right)^{\frac{1}{2}} \sin(kz) (a + a^\dagger), \tag{27}$$

with $\mathbf{D} = D \mathbf{n}$. The dipole operator, D , couples the fundamental and excited state, and it can be described as $D = d |g\rangle\langle e| + d^* |e\rangle\langle g|$. Note that we are only taking into consideration the transition between the two levels. If we assume that d is real, then $D = d\sigma_x$. In such a case, we can express the interaction Hamiltonian as

$$H_{\text{Inter}} = g\sigma_x (a + a^\dagger), \tag{28}$$

where the factor g contains all the constants³. We will refer to g as the coupling strength. We have gone through a detailed discussion of each term from the Rabi Hamiltonian.

³Note that the coupling strength g introduced here carries dimensions of frequency, unlike the dimensionless parameter g discussed in Subsection 4.1.

Now it is propitious to put all the pieces together, yielding

$$H_{\text{Rabi}} = \omega a^\dagger a + \Delta \sigma_z + g \sigma_x (a + a^\dagger). \quad (29)$$

With the exact mathematical description of our single-atom, single-mode system now established, we have reached the core of our metrological endeavor. However, merely possessing the Hamiltonian is not sufficient; to evaluate the fundamental precision limits using the QFI, we must understand the structure of the system's GS. This pursuit is the central focus of the following section.

6 Equations on the Edge: Probing Critical Metrology

Having formally derived the QRM, we are now fully equipped to apply the metrological concepts introduced in previous chapters and investigate the diverging QFI predicted near quantum phase transitions. Our primary objective in this section is to evaluate the asymptotic precision in the estimation of the coupling strength g . Note that we could have chosen to study the precision bounds for any other parameter embedded in the Hamiltonian—such as the qubit energy Δ , as proposed by Garbe et al. in Ref. [2].

We will compute the QFI for the GS family ϱ_g with respect to g and demonstrate that it formally diverges as the system approaches the critical point. Simultaneously, this analytical procedure will allow us to pinpoint the exact location of the QCP itself. Because the evaluation of the QFI fundamentally relies on the state vector, determining the GS of the system across the different physical regimes is our absolute priority. We will tackle this task both analytically and numerically.

Regarding the analytical approach, it is imperative to remark that the QRM was analytically solved in its entirety by D. Braak [26]. However, his exact solution relies on an exceptionally complex mathematical framework that is not easily adaptable for extracting simple, closed-form metrological bounds. Therefore, we will follow the standard and highly effective approach utilized in critical metrology [2, 16] and treat the problem approximately.

6.1 First contact: QFI for bosonic mean-field Hamiltonian

Our first approximation is going to involve a bosonic mean-field. The bosonic operators are expressed as $a \rightarrow \langle a \rangle + \delta a$ and $a^\dagger \rightarrow \langle a^\dagger \rangle + \delta a^\dagger$; there is a mean-field, and the dynamics are governed by fluctuations around that value. If we neglect the fluctuating term, the approximation consists of substituting the operator for its mean value, as was done for the Dicke model in Ref. [27]. We will denote $\langle a \rangle$ as α and $\langle a^\dagger \rangle$ as α^* , where $*$ means complex conjugate.

The approximated spin mean-field Hamiltonian becomes

$$H_{\text{SMF}} = \Delta \sigma_z + g \sigma_x (\alpha + \alpha^*). \quad (30)$$

Bear in mind that we dropped the factor $a^\dagger a$ as it is a constant in the mean field approximation. Certainly, $\alpha + \alpha^* = 2\text{Re}(\alpha)$, which we shall denote by A . So, the version of the Hamiltonian that we will use in the analysis is

$$H_{\text{SMF}} = \Delta \sigma_z + A g \sigma_x. \quad (31)$$

This Hamiltonian is purely spin-dependent and can be treated as a two-dimensional exercise. To compute the QFI via Eq. (15), we first need to find the GS of the system. For the two-level system, we will use the orthonormal basis corresponding to the spin-up and spin-down states, $\{|\uparrow\rangle = (1, 0)^T, |\downarrow\rangle = (0, 1)^T\}$.

In the abovementioned basis, σ_z, σ_x take on the following form

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (32)$$

The eigenvalues obtained upon diagonalization are $E_{\pm} = \pm\sqrt{A^2g^2 + \Delta^2}$. Now we are bound to find the eigenstate corresponding to the minimum energy. We will denote $E_- = -\sqrt{A^2g^2 + \Delta^2} = -R$. Now, the normalized GS takes on the form

$$|\text{GS}\rangle = \frac{1}{\sqrt{2R(R + \Delta)}} [Ag |\uparrow\rangle - (\Delta + R) |\downarrow\rangle], \quad (33)$$

which can be expressed more succinctly using a handy variable θ , such that $\cos \theta = -\frac{\Delta}{R}$ and $\sin \theta = \frac{Ag}{R}$. From these two follows that $\theta = \pi - \arctan\left(\frac{Ag}{\Delta}\right)$. We have successfully transformed the GS into

$$|\text{GS}\rangle = \cos \frac{\theta}{2} |\uparrow\rangle - \sin \frac{\theta}{2} |\downarrow\rangle. \quad (34)$$

We are ready to calculate the QFI applying Eq. (15). First, we compute $\partial_g |\text{GS}\rangle$

$$\partial_g |\text{GS}(\theta)\rangle = \partial_\theta |\text{GS}\rangle \frac{\partial \theta}{\partial g}. \quad (35)$$

Evaluating it step by step we find that $\partial_\theta |\text{GS}\rangle = \frac{1}{2} [-\sin \frac{\theta}{2} |\uparrow\rangle - \cos \frac{\theta}{2} |\downarrow\rangle]$ and $\theta'(g) = -\frac{A\Delta}{\Delta^2 + A^2g^2}$. Putting everything together into Eq. (15) we get

$$F_Q(\varrho_g) = \frac{A^2\Delta^2}{(\Delta^2 + A^2g^2)^2}. \quad (36)$$

We must make some safety checks on the result we obtained. The first is to see what occurs when $\Delta = 0$. If this is the case, the eigenstates do not depend on g , and therefore no information about g can be obtained from that Hamiltonian. We must also consider whether the dimensionality of the expression is adequate. We know for a fact that the QFI must have the dimensionality of the inverse square of the parameter that is to be estimated. In our case, $[F_Q(\varrho_g)] = 1/[g]^2$ must hold, and our result successfully does so. This result signals a non-diverging QFI, and does not reflect criticality; we must refine the analytical model.

6.2 Refined Effective Models: First- and Second-Order SW Transformation

This section aims to set the foundation to derive an approximate effective Hamiltonian and enable computation of the QFI. To achieve this, we will employ the first- and second-order SW transformation, a unitary frame change. We begin by reviewing the underlying theory and the mathematical implications of this transformation.

6.2.1 Fundamentals and Unitary Construction

The SW transformation is rigorously a unitary transformation method used to decouple weakly interacting subspaces to obtain a low-energy effective Hamiltonian [28]. The main objective is to transform the Hamiltonian into a block-diagonal (even) form. We begin with a Hamiltonian partitioned as,

$$H = H_0 + \varepsilon H_1, \quad (37)$$

where H_0 represents the unperturbed, block-diagonal (even) part of the Hamiltonian, while the perturbation H_1 is block-antidiagonal (odd). In the following derivations, the terms even/odd and block-diagonal/block-antidiagonal will be used interchangeably.

To construct the transformation, we define a unitary operator $U = e^S$, where the generator S is anti-Hermitian ($S^\dagger = -S$) and strictly odd. The transformed Hamiltonian is thus defined via,

$$U = e^S \rightarrow U^\dagger = e^{-S}, \quad (38)$$

which guarantees the required unitarity. The effective Hamiltonian takes the form:

$$H_{\text{eff}} = e^S H e^{-S}. \quad (39)$$

To expand this expression, we invoke the Baker–Campbell–Hausdorff (BCH) formula:

$$H_{\text{eff}} = e^S H e^{-S} \approx H + [S, H] + \frac{1}{2}[S, [S, H]] + \frac{1}{6}[S, [S, [S, H]]] + \frac{1}{24}[S, [S, [S, [S, H]]]] + \dots \quad (40)$$

This formula follows directly from the Taylor series expansion of the exponentials and applies to any operator.

The generator S is formally expanded in powers of ε as [28].

$$S = \sum_{k=1}^{\infty} \varepsilon^k s_k = \varepsilon^1 s_1 + \varepsilon^2 s_2 + \varepsilon^3 s_3 + \dots \quad (41)$$

where every s_k must be strictly odd. Note that ε is treated as a formal bookkeeping parameter to track the expansion order; it will be set to $\varepsilon = 1$ at the end of the calculation.

Introducing this expansion alongside Eq. (37) into the BCH formula (Eq. (40)) yields,

$$\begin{aligned} H_{\text{eff}} = & \varepsilon^0 H_0 + \varepsilon^1 (H_1 + [s_1, H_0]) \\ & + \varepsilon^2 ([s_2, H_0] + [s_1, H_1] + \frac{1}{2}[s_1, [s_1, H_0]]) \\ & + \varepsilon^3 ([s_3, H_0] + [s_2, H_1] + \frac{1}{2}([s_2, [s_1, H_0]] + [s_1, [s_2, H_0]] + [s_1, [s_1, H_1]])) \\ & \quad + \frac{1}{6}[s_1, [s_1, [s_1, H_0]]]) \\ & + \varepsilon^4 ([s_4, H_0] + [s_3, H_1] \\ & \quad + \frac{1}{2}([s_3, [s_1, H_0]] + [s_2, [s_2, H_0]] + [s_2, [s_1, H_1]] + [s_1, [s_3, H_0]] + [s_1, [s_2, H_1]])) \\ & \quad + \frac{1}{6}([s_2, [s_1, [s_1, H_0]]] + [s_1, [s_2, [s_1, H_0]]] + [s_1, [s_1, [s_2, H_0]]] + [s_1, [s_1, [s_1, H_1]]]) \\ & \quad + \frac{1}{24}[s_1, [s_1, [s_1, [s_1, H_0]]]]) \\ & + \mathcal{O}(\varepsilon^5). \end{aligned} \quad (42)$$

The objective is to systematically eliminate all odd terms from Eq. (42) by enforcing the odd contribution for each power of ε to vanish. This process relies on the commutator algebra for even ($\mathcal{E}$) and odd ($\mathcal{O}$) operators (detailed in Ref. [6]),

$$[\mathcal{E}, \mathcal{E}] = \mathcal{E}, \quad [\mathcal{E}, \mathcal{O}] = \mathcal{O}, \quad [\mathcal{O}, \mathcal{O}] = \mathcal{E}. \quad (43)$$

For our analysis, the approximation will be truncated at the first and second non-vanishing even corrections. With this algebraic framework established, we proceed to derive the specific constraints for each order.

6.2.2 First-Order Correction Constraints

To isolate the first-order correction, it suffices to truncate the expansion at ε^2 ,

$$H_{\text{eff}} = H_0 + \varepsilon(H_1 + [s_1, H_0]) + \varepsilon^2 \left([s_2, H_0] + [s_1, H_1] + \frac{1}{2}[s_1, [s_1, H_0]] \right) + \mathcal{O}(\varepsilon^3), \quad (44)$$

The generators s_1 and s_2 are determined by sequentially cancelling the odd terms at each order of ε . At order ε^0 , the term H_0 is inherently even. At order ε^1 , both H_1 and $[s_1, H_0]$ are odd. Nullifying this contribution yields the fundamental condition,

$$[s_1, H_0] = -H_1. \quad (45)$$

This relationship uniquely dictates the form of s_1 , which will be explicitly solved in the subsequent section.

At order ε^2 , the only odd component is $[s_2, H_0]$. Enforcing this term to vanish ($[s_2, H_0] = 0$), given that s_2 is strictly odd and H_0 is strictly even, requires $s_2 = 0$. This property extends generally: all even-indexed generators s_{2k} are identically zero [28].

Substituting Eq. (45) and $s_2 = 0$ into Eq. (44) produces the effective first-order Hamiltonian,

$$H'_{\text{SW1}} = H_0 + \frac{\varepsilon^2}{2}[s_1, H_1]. \quad (46)$$

As previously stated, the formal parameter ε will be set to 1 for all physical calculations.

6.2.3 Second-Order Correction Constraints

Incorporating the derived constraints (s_1 via Eq. (45) and $s_2 = 0$) into the general expansion (Eq. (42)) yields,

$$H_{\text{eff}} = H_0 + \frac{\varepsilon^2}{2}[s_1, H_1] + \varepsilon^3([s_3, H_0] + \frac{1}{3}[s_1, [s_1, H_1]]) + \varepsilon^4([s_4, H_0] + \frac{1}{2}[s_3, H_1] + \frac{1}{8}[s_1, [s_1, [s_1, H_1]]]) + \mathcal{O}(\varepsilon^5). \quad (47)$$

Having ensured that all terms up to ε^2 are purely even, the next odd contribution appears at order ε^3 , specifically $[s_3, H_0]$ and $\frac{1}{3}[s_1, [s_1, H_1]]$. Nullifying this sum establishes the constraint for the third-order generator as

$$[s_3, H_0] = -\frac{1}{3}[s_1, [s_1, H_1]]. \quad (48)$$

This condition dictates the form of s_3 . Proceeding to the ε^4 terms, all components are even except for $[s_4, H_0]$. Consistent with the prior deduction for even-indexed generators, this requires $s_4 = 0$. Consequently, the simplified effective second-order Hamiltonian reduces to,

$$H'_{\text{SW2}} = H_0 + \frac{\varepsilon^2}{2} [s_1, H_1] + \varepsilon^4 \left(\frac{1}{2} [s_3, H_1] + \frac{1}{8} [s_1, [s_1, [s_1, H_1]]] \right) + \mathcal{O}(\varepsilon^5). \quad (49)$$

With the underlying theoretical framework of the SW transformation established, we will now explicitly compute the first- and second-order effective Hamiltonians to subsequently evaluate the QFI.

6.3 First order SW transformed Hamiltonian

We must find the generator s_1 . There are several ways to go through this; we will find an appropriate generator by using the following steps. First, we need to define H_0 and H_1 . We take $H_0 = \omega a^\dagger a + \Delta \sigma_z$ and $H_1 = g \sigma_x (a + a^\dagger)$. There are other options, for instance, including the corotating terms (e.g., $\sigma_+ a$) into H_0 and leaving the counterrotating ones in H_1 .

With the selection we have made in mind, we recall Eq. (45). From the functional form of H_0 , the only possible way to obtain $-H_1$ from a commutator is that s_1 has the following form $s_1 = \sigma_y (\alpha a + \beta a^\dagger) + \sigma_x (C a + D a^\dagger)$, where α, β, C and D are parameters that we are about to find. This is the ansatz that will take us to an appropriate generator. We compute $[s_1, H_0]$:

$$\begin{aligned} [s_1, H_0] &= \sigma_y [\omega(-\alpha a + \beta a^\dagger) + 2i\Delta(C a + D a^\dagger)] \\ &\quad + \sigma_x [\omega(-C a + D a^\dagger) - 2i\Delta(\alpha a + \beta a^\dagger)], \end{aligned} \quad (50)$$

where we have essentially used that $[a^\dagger a, a] = -a$, $[a^\dagger a, a^\dagger] = a^\dagger$ and $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$, where ϵ_{ijk} is the Levi-Civita symbol. Now, solving for α, β, C and D yields

$$\alpha = \beta = \frac{2ig\Delta}{4\Delta^2 - \omega^2}, \quad C = \frac{g\omega}{4\Delta^2 - \omega^2}, \quad D = -\frac{g\omega}{4\Delta^2 - \omega^2}. \quad (51)$$

So, we end up with

$$s_1 = \frac{2ig\Delta}{4\Delta^2 - \omega^2} \sigma_y (a + a^\dagger) + \frac{g\omega}{4\Delta^2 - \omega^2} \sigma_x (a - a^\dagger). \quad (52)$$

This would be a great choice for the general QRM case; however, we must remember that it is the classical oscillator limit that concerns us. If we consider that $\omega/\Delta \rightarrow 0$ we end up with

$$s_1 = i \frac{g}{2\Delta} (a^\dagger + a) \sigma_y, \quad (53)$$

which is clearly anti-Hermitian owing to the imaginary unit. Moreover, it resembles the generator utilized by Garbe et al. in Ref. [2]. With s_1 at our disposal, we set out to get the transformed Hamiltonian.

Making use of basic commutator algebra, considering that operators from different Hilbert spaces commute and harnessing the indispensable commutation relationship

between the Pauli matrices, we obtain $[s_1, H_1] = \frac{g^2}{\Delta}(a + a^\dagger)^2 \sigma_z$, which ultimately leads from Eq. (46) to

$$H'_{\text{SW1}} = \omega a^\dagger a + \Delta \sigma_z + \frac{g^2}{2\Delta}(a + a^\dagger)^2 \sigma_z. \quad (54)$$

We have accomplished our first goal, deriving the first-order SW transformed Hamiltonian. Instead of rushing to obtain the exact GS of this Hamiltonian, we are first going to prove that the QCP of this Hamiltonian—and the QRM—is located at $g_c = \sqrt{\Delta\omega/2}$ ⁴.

6.3.1 Locating the QCP

In principle, an exact unitary transformation preserves the location of the QCP [29]. Having established the validity of the SW1 approximation at the critical point, we proceed to locate the QCP by analyzing the stability of the GS energy landscape. To target the lowest energy configuration, we project the Hamiltonian onto the lower spin eigenspace and then apply a bosonic mean-field ansatz. Assuming the mean-field amplitude α to be real, the effective GS energy as a function of the macroscopic bosonic occupation number $|\alpha|^2 = \langle a^\dagger a \rangle$ is given by:

$$E_I(|\alpha|^2) = -\Delta + \left(\omega - \frac{2g^2}{\Delta} \right) |\alpha|^2. \quad (55)$$

The location of the quantum phase transition is determined by the coefficient of the effective potential, $\frac{\partial E_I}{\partial |\alpha|^2}$. The critical coupling g_c is identified as the exact point where this coefficient vanishes,

$$g_c = \sqrt{\frac{\Delta\omega}{2}}, \quad (56)$$

while keeping the scaled parameter $\gamma = g/\sqrt{\Delta\omega}$ finite. At this precise threshold, the stability of the GS undergoes a drastic change. For $g < g_c$, the energy increases monotonically with the photon number, ensuring the vacuum state ($|\alpha|^2 = 0$) remains a stable global minimum. Conversely, for $g > g_c$, the coefficient becomes negative. In this regime, the GS energy decreases unboundedly as photons are added, signaling a continuous transition into a superradiant phase. This unphysical, unbounded energy landscape indicates that the truncated SW1 treatment breaks down beyond the critical point—a limitation we will verify analytically and explore further in the subsequent numerical analysis. With the critical point rigorously located, it is timely to obtain the exact GS of the SW1 transformed Hamiltonian within its domain of validity.

6.3.2 Exact GS of SW1 transformed Hamiltonian

So, why has applying a SW transformation turned out helpful? Simply because the Hamiltonian is diagonal in the spin subspace. By projection in the lower spin eigenspace, we are minimizing the energy (getting closer to our desired GS). We end up with a purely bosonic Hamiltonian,

$$H_{\text{np}} = \langle \downarrow | H'_{\text{SW1}} | \downarrow \rangle = \omega a^\dagger a - \Delta - \frac{g^2}{2\Delta}(a + a^\dagger)^2. \quad (57)$$

⁴This value is consistent with that obtained by Garbe et al. in Ref. [2] given the notation utilized. The Pauli matrices they used have a prefactor 1/2, hence that factor 2.

This quadratic bosonic Hamiltonian can be brought to a harmonic oscillator form; we hereby invoke the Bogoliubov transformation. For the mathematical details, see Appendix A. Conceptually, this method consists of a canonical transformation that mixes the original creation and annihilation operators to define a new set of them. By expressing the Hamiltonian in terms of these new normal modes, the GS of the system can be straightforwardly identified as the vacuum state of the transformed operators [25].

To obtain the required form of the Hamiltonian (see Appendix A), we first reorder the Hamiltonian as

$$H_{\text{np}} = (\omega - \frac{g^2}{\Delta})a^\dagger a - \frac{g^2}{2\Delta}(aa + a^\dagger a^\dagger) - \Delta - \frac{g^2}{2\Delta}. \quad (58)$$

From this, we extract that $\xi = \omega - \frac{g^2}{\Delta}$, and $\kappa = -\frac{g^2}{2\Delta}$. Notice that since κ is negative, that is $\phi = \pi$, we choose $\theta = \pi$. Making use of Appendix A, we find the modulus of the squeezing parameter z as

$$r \equiv |z| = -\frac{1}{4} \ln \left(1 - \frac{2g^2}{\omega\Delta} \right). \quad (59)$$

A little intermission is due, because this result predicts the location of the QCP we forecasted. The argument of Eq. (59) becomes zero at $g_c = \sqrt{\frac{\omega\Delta}{2}}$. Not only that, the argument becomes negative in the superradiant phase, which is inconsistent with r being real, pointing out that this treatment is only applicable in the normal phase.

Returning to our main objective, we must now obtain an approximate GS vector of our initial QRM Hamiltonian. From the Bogoliubov analysis we just committed, we know that the GS of $H_{\text{SW}1}$ is $|\text{GS}\rangle = S_z |0\rangle \otimes |\downarrow\rangle$, where S_z is the squeezing operator (see Appendix A). So, the exact GS energy for this Hamiltonian is

$$E_{\text{GS}} = \langle \downarrow | \otimes \langle 0 | S_z^\dagger H_{\text{SW}1} S_z | 0 \rangle \otimes |\downarrow\rangle. \quad (60)$$

We know that the relationship between H_{QRM} and the full SW transformed Hamiltonian is $H_{\text{SW}} = e^S H_{\text{QRM}} e^{-S}$. Of course, we have not applied the whole expansion, as we have only accounted for the first non-null even correction. Therefore, we may take the following approximation $H_{\text{SW}1} \approx e^{s_1} H_{\text{QRM}} e^{-s_1}$ (which is an approximation itself). So, inserting this into Eq. (60) yields

$$E_{\text{GS}} = \langle \downarrow | \otimes \langle 0 | S_z^\dagger e^{s_1} H_{\text{QRM}} e^{-s_1} S_z | 0 \rangle \otimes |\downarrow\rangle. \quad (61)$$

So the approximate GS for the QRM is:

$$|\psi_{\text{probe}}\rangle = e^{-s_1} S_z |0\rangle \otimes |\downarrow\rangle. \quad (62)$$

This expression closely resembles a variational ansatz for the ground state. However, a purely variational method fails here, as determining the optimal parameters z and s_1 analytically proves impossible. Instead, our approach uses the Schrieffer-Wolff transformation and the subsequent Bogoliubov diagonalization to systematically derive these energy-minimizing parameters. As a consistency check, setting $g = 0$ seamlessly recovers the uncoupled ground state $|\text{GS}_{\text{QRM}}\rangle = |0, \downarrow\rangle$, as the expressions for s_1 and $r(g)$ readily demonstrate.

6.3.3 Computation of QFI

We compute the QFI using Eq. (15). In this section, we will only give a hint at the calculation due to its extension. The full detailed derivation can be found in Ref. [6]. The calculation simplifies significantly due to symmetry. The overlap term $\langle \psi | \partial_g \psi \rangle$ vanishes identically because the derivative operators $-\sigma_y$ from the displacement and $(a^{\dagger 2} - a^2)$ from the squeezing—have zero expectation values in the GS. Consequently, the QFI reduces to the norm of the derivative vector, $4\langle \partial_g \psi | \partial_g \psi \rangle$. This vector decomposes into two orthogonal components associated with the displacement and squeezing variations. Evaluating their respective norms using standard squeezed-state algebra yields:

$$F_Q(\varrho_g) = \frac{e^{-2r}}{\Delta^2} + 2(\partial_g r)^2. \quad (63)$$

Substituting the analytical expressions for $r(g)$ and $\partial_g r$ derived previously, we arrive at the final result for the normal phase:

$$F_Q(\varrho_g) = \frac{\sqrt{1 - \frac{2g^2}{\omega\Delta}}}{\Delta^2} + \frac{2g^2}{(\omega\Delta - 2g^2)^2}. \quad (64)$$

This expression confirms the metrological advantage of the system, exhibiting a divergence as the coupling approaches the critical point g_c . Applying the CRB means that the coupling strength could, in principle, be estimated with 0 error (variance).

Regarding how the QFI behaves near the critical point, expanding the second term about the critical point enables us to write it as $F_Q(\varrho_g) \approx \frac{1}{8}(g_c - g)^{-2}[1 + \mathcal{O}(\frac{\delta g}{g})]$, yielding a critical exponent of -2 . The same happens if we use γ instead of g .

6.4 Second order SW transformed Hamiltonian

Our first mission here is to find an s_3 that fulfills Eq. (48). To get to that, we first compute $[s_1, [s_1, H_1]]$. We have a head start as we have already calculated $[s_1, H_1]$. We find that

$$[s_1, [s_1, H_1]] = -\frac{g^3}{\Delta^2}(a + a^\dagger)^3 \sigma_x, \quad (65)$$

So, we want to find s_3 such that $[s_3, H_0] = \frac{g^3}{3\Delta^2}(a + a^\dagger)^3 \sigma_x$. As we did for the first order, we propose a general form for s_3

$$s_3 = i(F\sigma_y + G\sigma_x), \quad (66)$$

where F and G are Hermitian bosonic operators (they commute with the Pauli matrices), so that s_3 is anti-Hermitian. This case is not straightforward, and we must find F and G as we go. We start by obtaining $[s_3, H_0]$. For the sake of simplicity, we will denote $\mathcal{A} = a + a^\dagger$ and introduce the number operator $N = a^\dagger a$. So, we have that,

$$[s_3, H_0] = \sigma_x(i\omega[G, N] - 2\Delta F) + \sigma_y(i\omega[F, N] + 2\Delta G) = \frac{g^3}{3\Delta^2}\mathcal{A}^3\sigma_x. \quad (67)$$

We extract two equations from this,

$$\begin{aligned} G &= -\frac{i\omega}{2\Delta}[F, N] \\ i\omega[G, N] - 2\Delta F &= \frac{g^3}{3\Delta^2}\mathcal{A}^3. \end{aligned} \quad (68)$$

To solve this, we simply remember that we are considering the classical oscillator limit of the QRM. That being the case ($\omega/\Delta \rightarrow 0$) and owing to the fact that $[F, N]$ will not contain any Δ or ω , we can take $G = 0$. Consequently, we easily solve for F

$$F = -\frac{g^3}{6\Delta^3} \mathcal{A}^3, \quad (69)$$

which if introduced into Eq. (66) yields an antiHermitian operator,

$$s_3 = -\frac{ig^3}{6\Delta^3} (a + a^\dagger)^3 \sigma_y. \quad (70)$$

We are finally ready to obtain the second-order SW effective Hamiltonian using Eq. (49). We arrive at

$$H'_{\text{SW2}} = \omega a^\dagger a + \Delta \sigma_z + \frac{g^2}{2\Delta} (a + a^\dagger)^2 \sigma_z - \frac{7g^4}{24\Delta^3} (a + a^\dagger)^4 \sigma_z. \quad (71)$$

Unfortunately, applying a Bogoliubov transformation is not feasible for a quartic Hamiltonian like this one. Our approach to the problem is to use a beyond mean field approximation where we only keep quadratic terms—we eliminate the linear term to find the minimum energy and *neglect* the cubic and quartic—with the ultimate goal of enabling a Bogoliubov. This is our hypothesis. Therefore, we take

$$a \rightarrow \alpha + \delta a \text{ and } a^\dagger \rightarrow \alpha^* + \delta a^\dagger \quad (72)$$

The commutators of δa and $\delta a^\dagger$ are inherited from a and $a^\dagger$. The quadratic and linear components (upon projection in the lower spin eigenspace to minimize energy) are

$$H_{\text{linear}} = \omega(\alpha^* \delta a + \alpha \delta a^\dagger) - \frac{g^2}{\Delta} (\alpha + \alpha^*)(\delta a + \delta a^\dagger) + \frac{7g^4}{6\Delta^3} (\alpha + \alpha^*)^3 (\delta a + \delta a^\dagger), \quad (73)$$

$$H_{\text{quadratic}} = \omega \delta a^\dagger \delta a - \left(\frac{g^2}{2\Delta} - \frac{7g^4}{4\Delta^3} (\alpha + \alpha^*)^2 \right) (\delta a^2 + \delta a^{\dagger 2} + 2 \delta a^\dagger \delta a + 1). \quad (74)$$

We require that the linear term vanishes to find the mean-field.

$$\alpha \left[1 - \frac{2g^2}{\omega \Delta} + \frac{28g^4}{3\omega \Delta^3} \alpha^2 \right] = 0, \quad (75)$$

so the solutions are

$$\alpha = 0 \text{ and } \alpha^2 = \frac{3\omega \Delta^3}{-28g^4} \left(1 - \frac{2g^2}{\omega \Delta} \right). \quad (76)$$

For $g < g_c$, the second solution gives a negative α^2 , this has no physical meaning, as we chose α to be real. We discard that solution, simplifying to a null mean-field, $\alpha = 0$. The quadratic Hamiltonian after substituting $\alpha = 0$ is

$$H_{\text{quadratic}}^{\alpha=0} = \omega \delta a^\dagger \delta a - \frac{g^2}{2\Delta} (\delta a^2 + \delta a^{\dagger 2} + 2 \delta a^\dagger \delta a). \quad (77)$$

The first step towards a Bogoliubov transformation is to identify κ and ξ . We recover the same factors we had for the SW1 transformed Hamiltonian, that is $\xi = \omega - \frac{g^2}{\Delta}$ and $\kappa = -\frac{g^2}{2\Delta}$. This leads to the same squeezing parameter as before, and ultimately to the same GS for the transformed Hamiltonian. Now, to find an approximate GS for the full QRM, we do the same as before. The approximation comes from taking $H_{\text{SW2}} \approx e^{s_1 + s_3} H_{\text{QRM}} e^{-s_1 - s_3}$ (remember $s_2 = 0$). Our result for an approximate GS of the QRM is

$$|\psi_{\text{probe}}\rangle = e^{-s_1 - s_3} S^z |0\rangle \otimes |\downarrow\rangle. \quad (78)$$

6.4.1 Computation of QFI for the second order approximation

We extend the analysis to the GS obtained via the second-order SW transformation, as always using Eq. (15). The second order approximation introduces a cubic term in the anti-Hermitian generator, $s_3 \propto (a^\dagger + a)^3$, resulting in a derivative vector $\partial_g |\text{GS}_{\text{ORM}}\rangle$ composed of three distinct contributions (one more than before). This calculation is longer than the one for the SW1 Hamiltonian. Again, all the algebraic steps will be omitted; we will just give out a couple of hints. The full derivation can be accessed in the repository [6].

Symmetry arguments drastically simplify the evaluation of the general formula. However, unlike the first-order case, the linear and cubic displacement terms share the same spin sector and interfere. QFI calculation, therefore, requires the evaluation of their individual norms as well as their cross-term overlap. These terms are computed carefully. We finally arrive at

$$F_Q(\varrho_g) = \frac{4g^2}{\Delta^4} e^{-4r} + \frac{e^{-2r}}{\Delta^2} + \frac{20g^4}{3\Delta^6} e^{-6r} + 2(\partial_g r)^2. \quad (79)$$

Inserting the explicit functions for $r(g)$, we arrive at the final analytical result:

$$F_Q(\varrho_g) = \frac{4g^2}{\Delta^4} \left(1 - \frac{2g^2}{\Delta\omega}\right) + \frac{1}{\Delta^2} \sqrt{1 - \frac{2g^2}{\omega\Delta}} + \frac{20g^4}{3\Delta^6} \left(1 - \frac{2g^2}{\Delta\omega}\right)^{\frac{3}{2}} + \frac{2g^2}{(\omega\Delta - 2g^2)^2}. \quad (80)$$

This result retains the critical divergence observed in the first-order case while incorporating corrections to the non-critical zone.

6.5 What can we learn from these results?

We have obtained the QFI for both the first and second-order SW-transformed Hamiltonians. The next logical step, before delving into the numerical approach, is to interpret and compare these results. Both results have the same diverging term. We are going to study the regular term ($\mathcal{J}$); thus, we graph both as functions of $\gamma = g/\sqrt{\Delta\omega}$ —as per usual in the literature—with the QCP at $\gamma_c^2 = 1/2$. We get

$$\mathcal{J}^{\text{SW1}} = \frac{\sqrt{1 - 2\gamma^2}}{\Delta^2}, \quad (81)$$

$$\mathcal{J}^{\text{SW2}} = \frac{4\omega\gamma^2}{\Delta^3} (1 - 2\gamma^2) + \frac{\sqrt{1 - 2\gamma^2}}{\Delta^2} + \frac{20\omega^2\gamma^4}{3\Delta^4} (1 - 2\gamma^2)^{3/2}. \quad (82)$$

We fix $\Delta = 1$ and plot both expressions as a function of γ^2 . Taking fixed values of ω is very informative. The results are shown in Figure 3.

We see that making ω smaller, that is, approaching the classical oscillator limit, both results tend to overlap. This means that for the strict limit where the abrupt phase transition occurs, we have not obtained any new information from the second-order transformation. All the calculations we performed are, nonetheless, useful. They may not provide us with more insight regarding the phase transition, but we have attained a better characterization of the QFI for general ω and Δ values.

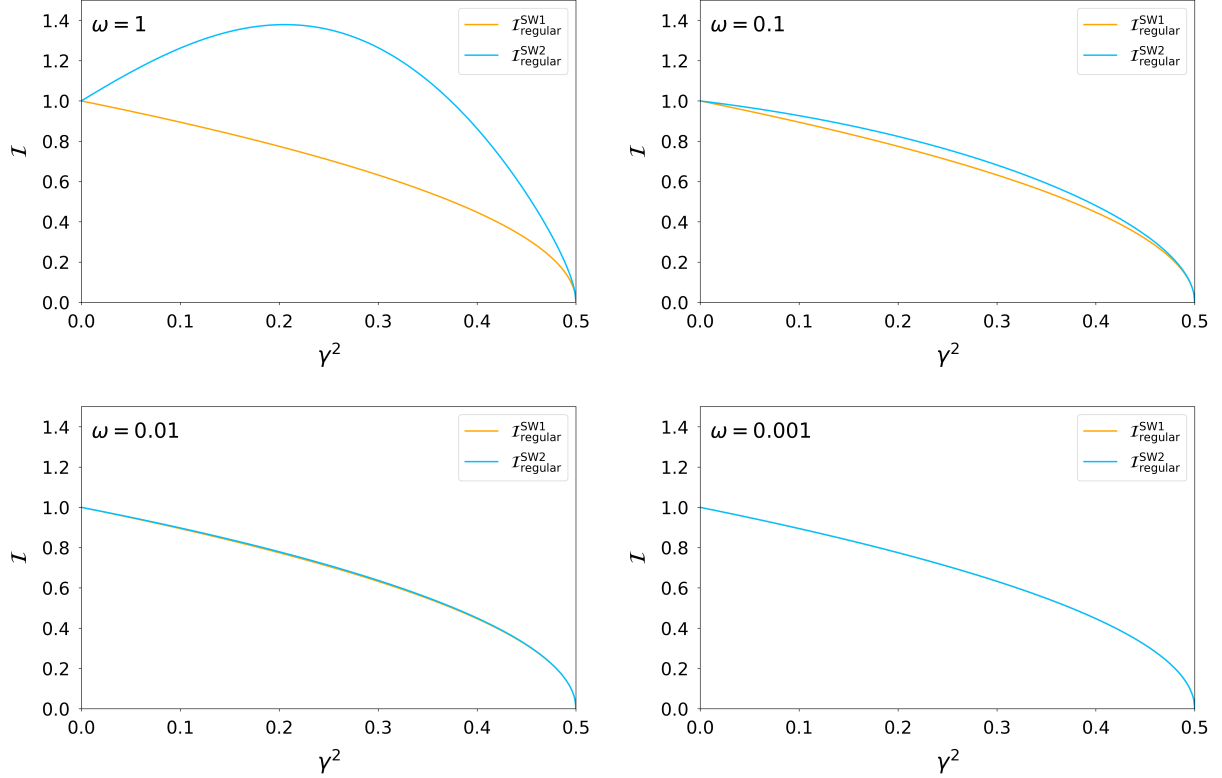


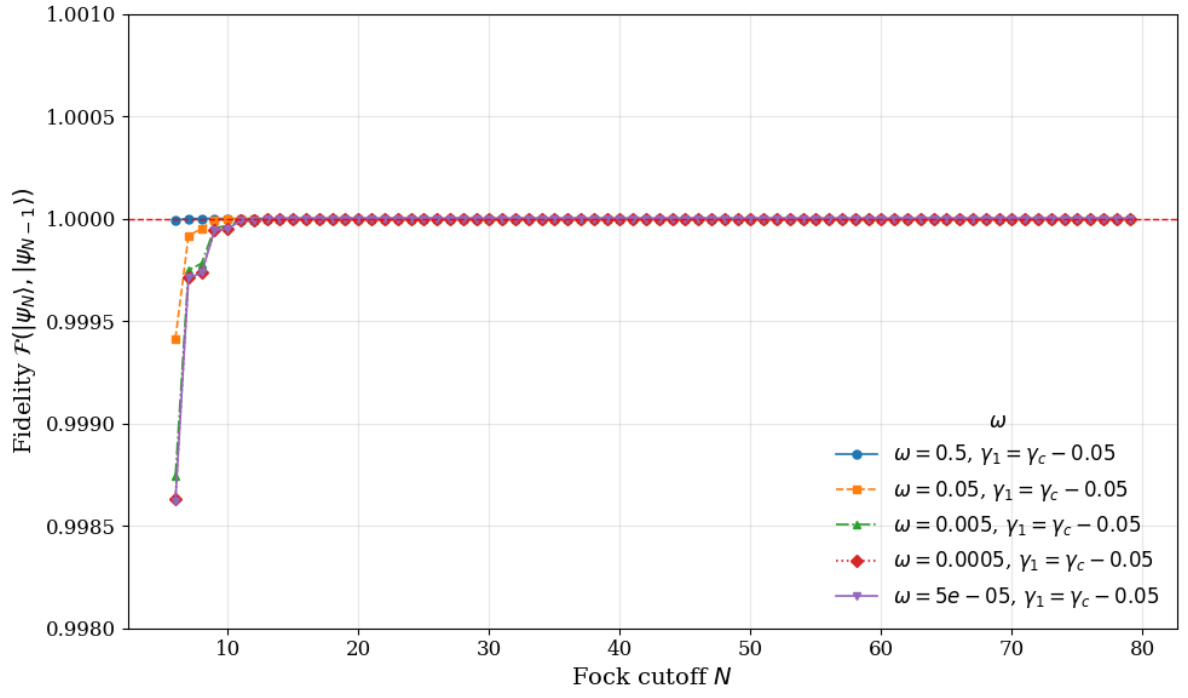
Figure 3: Regular part of the QFI obtained from the SW1 and SW2 transformed Hamiltonians with $\Delta = 1$. From left to right and top to bottom, the panels correspond to cavity frequencies $\omega = 1, 0.1, 0.01$, and 0.001 .

7 Numerical approach: Validation and further analysis

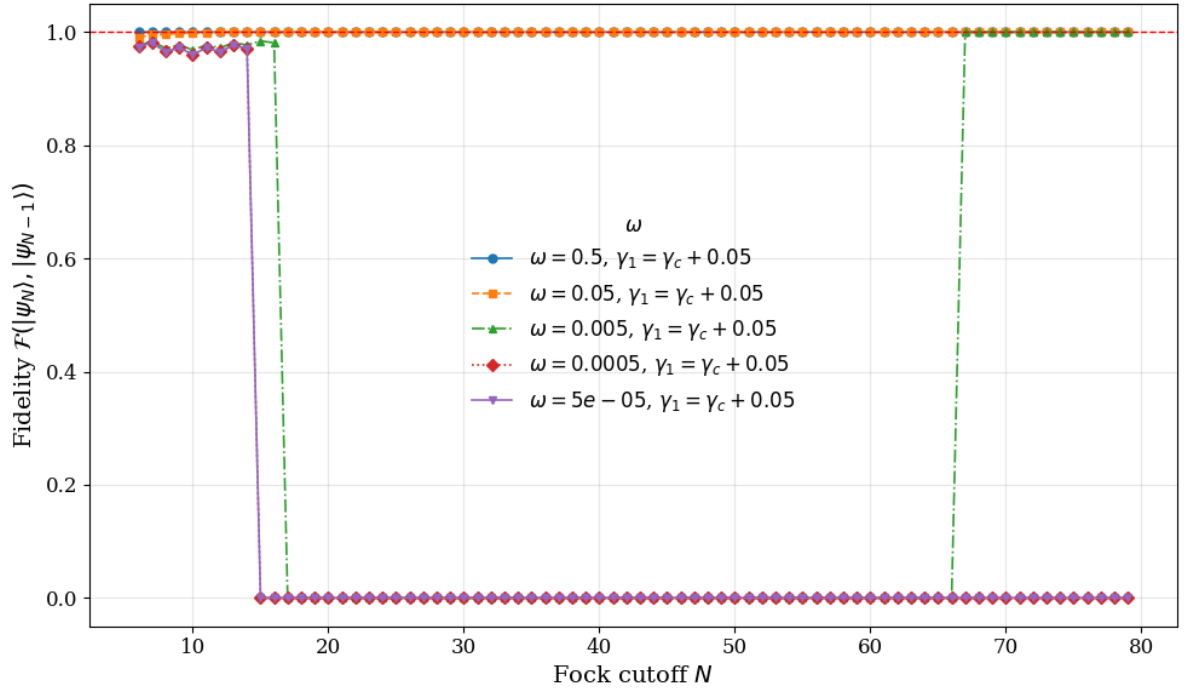
Thus far, we have tackled the problem analytically, utilizing both a fundamental mean-field approach and two different refined SW transformations. We now shift our attention to a numerical framework. The primary goal of this section is to numerically reproduce and validate our analytical predictions, specifically the divergence of the QFI near the critical point and the system's behavior across different parameter regimes.

Numerically, obtaining the GS requires diagonalizing the system's Hamiltonian. However, because the bosonic Hilbert space is inherently infinite-dimensional, a truncation must be applied. To ensure physical accuracy, it is imperative to verify whether the GS can strictly converge within a truncated subspace before extracting any quantities.

To analyze this convergence behavior (see Fig. 4) and evaluate how a standard truncation performs on each side of the transition, we adopt a specific test configuration. In line with the standard convention for numerical treatments of the QRM, we express our results in terms of the dimensionless normalized coupling γ instead of the bare coupling g . Rather than simulating the exact classical oscillator limit ($\omega/\Delta \rightarrow 0$)—which would trivially imply $g = 0$ for γ to remain finite—we fix $\Delta = 1$ and analyze varying values of ω approaching the limit. For this initial verification, we probe both regimes by fixing $\gamma = \gamma_c - 0.05$ for the normal phase and $\gamma = \gamma_c + 0.05$ for the superradiant phase. This parameter choice is intended solely to contrast the stability of the basis across the



(a)



(b)

Figure 4: Fidelity between consecutive truncations as a function of the cutoff for different ω values with $\Delta = 1$. (a) In the normal phase and (b) in the superradiant phase.

transition, rather than defining a general configuration for the rest of the numerical study.

To quantify the convergence under this setup, we evaluated the fidelity between GSs obtained from consecutively truncated Fock spaces, defined as $\mathcal{F} = |\langle \psi_N | \psi_{N-1} \rangle|^2$. The numerical computation of this overlap between spaces of different dimensions requires careful padding techniques, the details of which are documented in Ref. [6]. The resulting fidelities are exhibited in Figure 4.

Understanding Figure 4 is central to safely computing the QFI numerically. We clearly see that in the normal phase, a simple cutoff suffices (we will generally take $N = 60$ in upcoming simulations). The superradiant phase, however, is special; as expected, the sharp increase in bosonic occupation requires a prohibitively higher truncation. For this reason, we will initially deal only with the normal phase and shortly introduce specific analytical tricks of the trade to work in the superradiant zone.

7.1 Computation of QFI (normal phase)

Using numerical diagonalization, we can compute the QFI via Eq. (15). The numerical derivative must be treated carefully, specifically the choice of the finite step size, to minimize truncation and round-off errors. The detailed justification for the optimal step size is available in Ref. [6].

Having established that a Fock truncation of $N = 60$ guarantees convergence in the normal phase, we compute the QFI, as illustrated in Figure 5. We obtained a clear divergence in QFI when approaching the QCP.

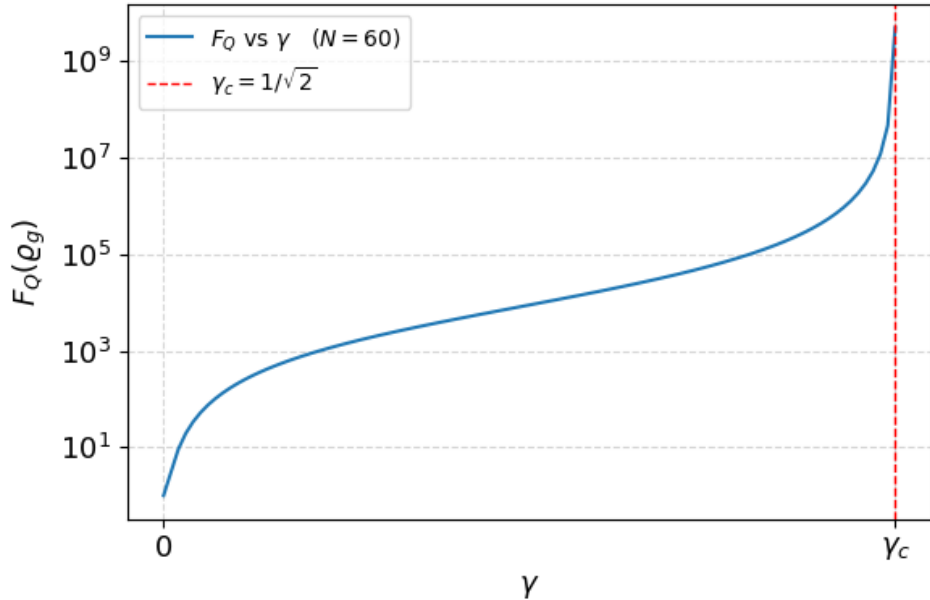


Figure 5: Numerical QFI in the normal phase for fixed $\omega/\Delta = 5 \cdot 10^{-5}$ and a reasonable cutoff. Using the full QRM Hamiltonian. Refer to Ref.[6] for the code.

We can explicitly study the critical exponent through this numerical result. Analytically, via the first-order SW transformation, we found that the diverging term of the QFI scales

as $\approx (\gamma_c - \gamma)^{-2}$ (see Eq. (64)), omitting non-diverging prefactors. Consistent with the findings of Garbe et al. [2], this scaling dictates a critical exponent of -2

By plotting the QFI as a function of the distance to the critical point ($\gamma_c - \gamma$) on a log-log scale, this power-law behavior should manifest as a straight line with a slope of -2 . This predicted tendency is clearly confirmed in the linear regime of Figure 6. However, in the immediate vicinity of the critical point, the numerical curve deviates from the theoretical divergence. This behavior is physically sound: our numerical simulation utilizes a finite ω/Δ ratio rather than the strict classical oscillator limit, and the system undergoes a smooth crossover rather than a sharp phase transition. Consequently, the true non-analyticity is regularized.

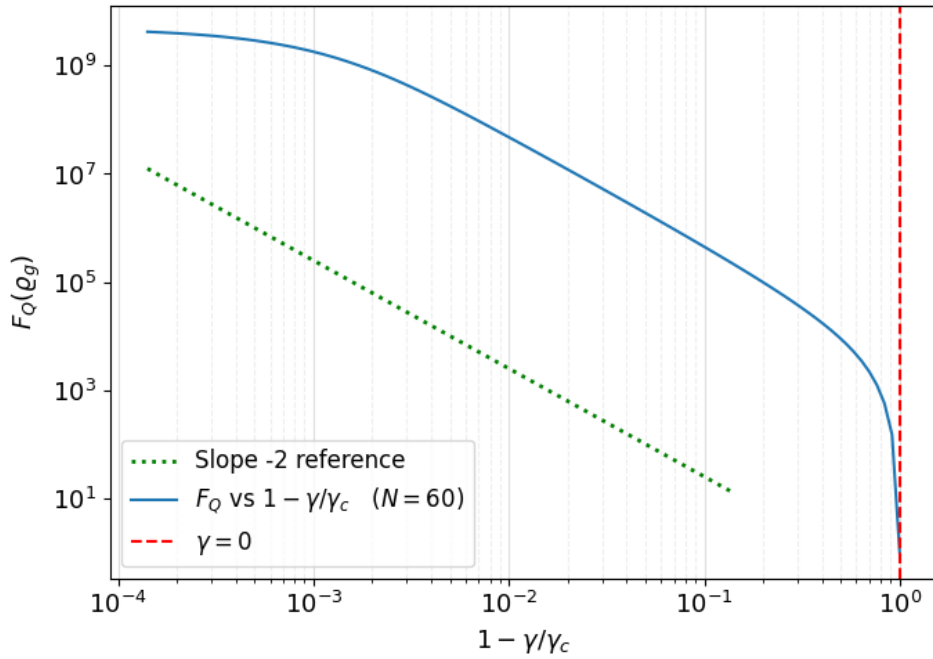


Figure 6: QFI vs $1 - \gamma/\gamma_c$ in a log-log plot compared to a straight line of slope -2 near the QCP. Refer to Ref. [6] for the code.

7.1.1 QFI representation via Heatmap

So far, we have only considered a single ω/Δ band to compute the QFI. We can extend to more values via a heatmap representation to grasp what happens outside the limit $\omega/\Delta \approx 0$.

Figure 7 displays a very important result: even if we are not in the classical oscillator limit, there is an enhancement in susceptibility when we get close to the critical point. Critical metrology is not restricted to a strict band of phase space; it can be applied to a broader regime. At least in the normal phase. We have to tackle the superradiant phase.

7.2 The Superradiant Phase: Displaced and Squeezed Bases

We now address the breakdown of the SW1 approximation in the superradiant phase. This failure can be intuitively understood by expressing Eq. (54) in terms of position and

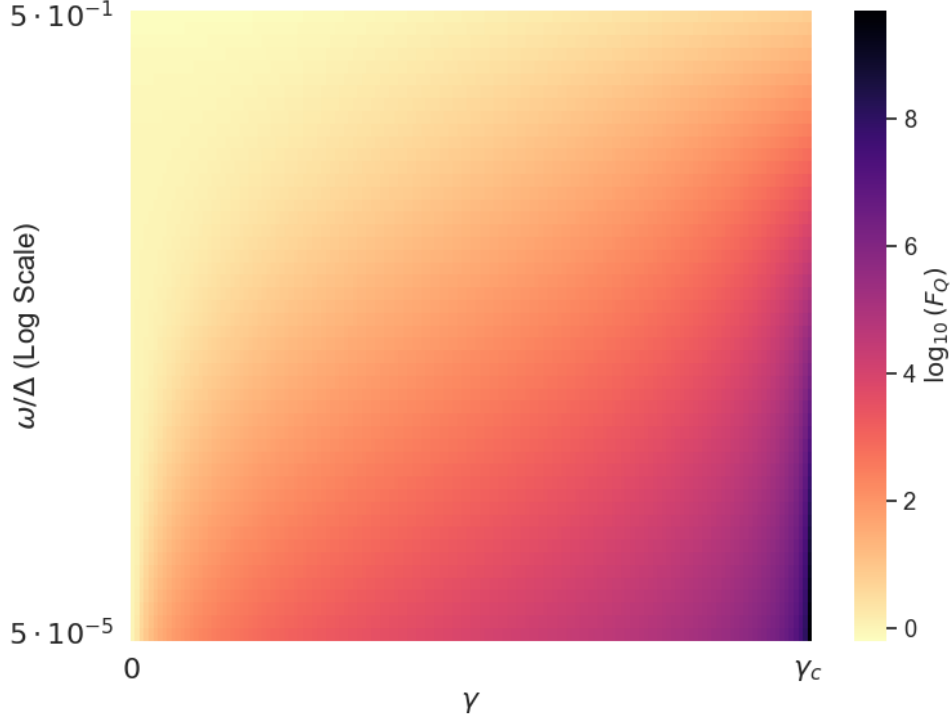


Figure 7: QFI heatmap in the normal phase, illustrating the enhancement of $\log_{10}(F_Q)$ near the critical point. 50 ω/Δ bands represented in a log scale. x-axis in linear scale. Code in Ref. [6].

momentum quadratures (after projection into the lower spin subspace),

$$H_b^{(\downarrow)} = \frac{\omega}{2} p^2 + \left(\frac{\omega}{2} - \frac{g^2}{\Delta} \right) x^2 + \text{constant}. \quad (83)$$

The coefficient of the effective harmonic potential is proportional to $(\gamma_c - \gamma)$. In the normal phase ($\gamma < \gamma_c$), the potential is a standard parabola with a stable minimum at $x = 0$. However, in the superradiant phase ($\gamma > \gamma_c$), the coefficient becomes negative, turning the potential into an inverted parabola where $x = 0$ is a local maximum. A standard Bogoliubov transformation is mathematically inapplicable to a potential with negative curvature. Consequently, the standard Fock basis centered at the origin becomes highly inefficient, as evidenced by the fidelity drop in Figure 4(b).

Taking this into consideration, we get back to the full QRM model. Following the treatment of Garbe et al. [2], we wish to apply a displacement to the Hamiltonian to locate the origin in a stable point. We first write the Hamiltonian as,

$$H_{\text{QRM}} = \Delta \left[\frac{a^\dagger a}{\Delta/\omega} + \sigma_z + \gamma \frac{a + a^\dagger}{\sqrt{\Delta/\omega}} \sigma_x \right]. \quad (84)$$

Now, we take a variational approximation to the GS with separable states $|\text{GS}\rangle = |\alpha\rangle \otimes |\psi\rangle$ where $|\alpha\rangle$ is a coherent state and $|\psi\rangle$ is a general spin state of the form $|\psi\rangle = \cos \theta |\uparrow\rangle + \sin \theta |\downarrow\rangle$. The variational GS energy $E_{\text{QRM}} = \langle \text{GS} | H_{\text{QRM}} | \text{GS} \rangle$ is

$$E_{\text{QRM}} = \Delta [x^2 + y^2 + \langle \psi | \sigma_z + 2\gamma x \sigma_x | \psi \rangle], \quad (85)$$

where we have taken $\alpha = (x + iy)\sqrt{\Delta/\omega}$ (mean field) with x and y real. Using the density operator formalism, we have that $\langle \psi | \sigma_z + 2\gamma x \sigma_x | \psi \rangle = \text{Tr}(|\psi\rangle \langle \psi| (\sigma_z + 2\gamma x \sigma_x)) = \cos 2\theta + 2\gamma x \sin 2\theta$. To make our life easier we can write this as $\cos 2\theta' \sqrt{1 + 4\gamma^2 x^2}$ where $\theta' = \theta - \phi/2$ and $\phi = \arctan 2\gamma x$. So we write the following,

$$E_{\text{QRM}} = \Delta[x^2 + y^2 + \cos 2\theta' \sqrt{1 + 4\gamma^2 x^2}]. \quad (86)$$

We have a function depending on three parameters and wish to minimize it for each of them. We trivially see that $\theta' = \pi/2$ and $y = 0$. So, the function to be minimized is simplified to $f(x) = x^2 - \sqrt{1 + 4\gamma^2 x^2}$ (graphed in Fig. 8). We look for the minima,

$$\frac{df(x)}{dx} = 0 = 2x - \frac{4\gamma^2 x}{\sqrt{1 + 4\gamma^2 x^2}}, \quad (87)$$

so $x = 0$ and $x^2 = \gamma^2 - \frac{1}{4\gamma^2}$ are our solutions. The second one makes no sense for the normal phase, as we have defined x to be real. In the superradiant phase, however, we find that $x = \pm\sqrt{\gamma^2 - \frac{1}{4\gamma^2}}$ are minima of the GS energy (and are symmetric), as we can see in Fig. 8.

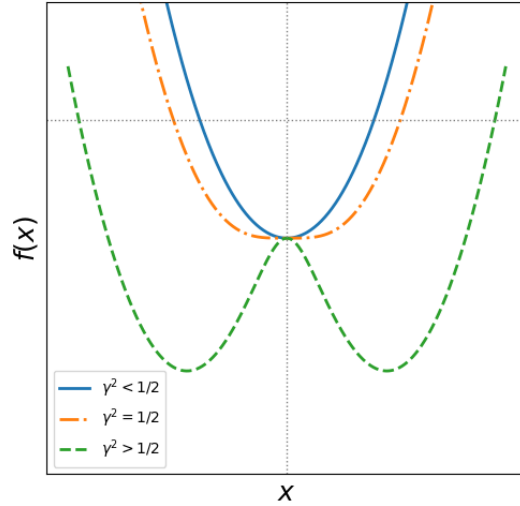


Figure 8: Visualization of the variational GS energy in the normal and superradiant phases. There is a spontaneous symmetry breaking.

Having found the positions of the minima, we set out to apply a displacement to the Hamiltonian (which physically accounts to shifting the origin). The displacement operator is defined as $D(\alpha) = \exp\{\alpha a^\dagger - \alpha^* a\}$ [2]. We have seen that displacements that bring about stable dynamics in the superradiant phase are $\alpha = \pm\alpha_s = \pm\sqrt{\gamma^2 - \frac{1}{4\gamma^2}}\sqrt{\frac{\Delta}{\omega}}$.

We apply that displacement to the QRM Hamiltonian. We know from Ref. [25] that

$$D^\dagger(\alpha) a D(\alpha) = a + \alpha \quad \text{and} \quad D^\dagger(\alpha) a^\dagger D(\alpha) = a^\dagger + \alpha^*.$$

With this in mind, the displaced Hamiltonian becomes

$$H_\pm = D^\dagger(\pm\alpha_s) H_{\text{Rabi}} D(\pm\alpha_s) = \omega a^\dagger a + \Delta \sigma_z \pm \omega \alpha_s (a + a^\dagger) + g \sigma_x (a + a^\dagger) \pm 2g \alpha_s \sigma_x + \omega \alpha_s^2. \quad (88)$$

Intuitively, applying a displacement operator will significantly enhance convergence in the superradiant regime by realigning the computational basis with the system's macroscopic mean field. Since the GS in this phase acquires a large average photon number, the standard vacuum-centered Fock basis requires a prohibitively high truncation to capture the state. The unitary displacement transformation shifts the phase space origin to the state's center ($|\langle a \rangle| \approx \alpha$), effectively reducing the problem to describing small quantum fluctuations around a classical field. This concentrates the probability amplitude within the lowest-lying Fock states of the transformed basis, allowing for accurate numerical diagonalization with a drastically reduced Hilbert space dimension, which ultimately eases numerical calculations.

With this, *a priori*, adequate Hamiltonian, we can readily compute the fidelities between consecutive bosonic truncations. Remember that the initial Hamiltonian yielded close to 0 fidelity for ω/Δ approaching the classical oscillator limit.

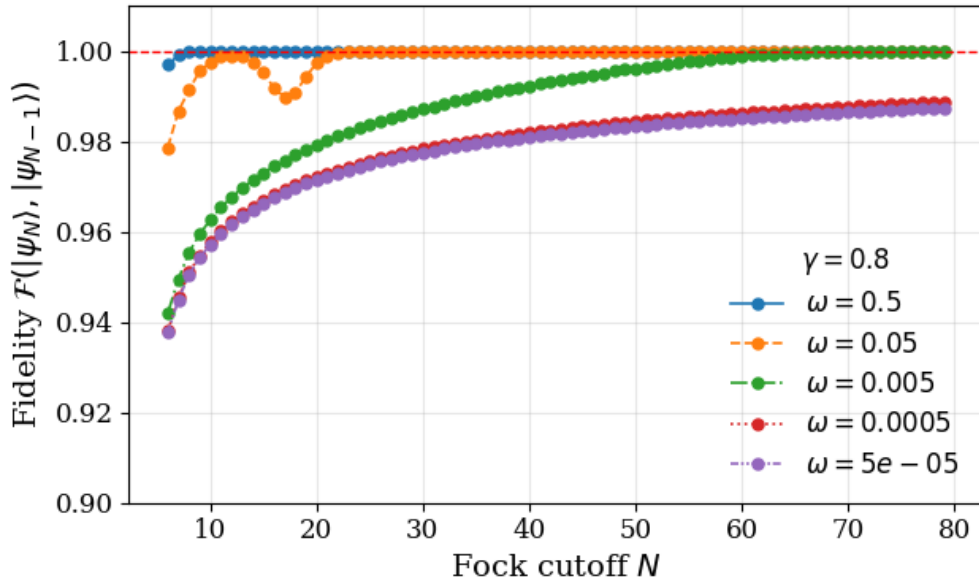


Figure 9: Fidelity between consecutive bosonic truncation in the superradiant phase with the displaced Hamiltonian. Refer to Ref. [6] for the code.

We clearly see from Fig. 9 that a reasonable bosonic cut-off results in adequate fidelities ($\mathcal{F} > 0.98$). We have made a significant improvement. We have also computed the QFI in the superradiant phase, studied its critical exponent, and generated the corresponding heatmap. While these figures are omitted here for brevity, the full code is available in Ref. [6]. We are going to devise a new method to improve the fidelities from Fig. 9. For that matter, we need to work out the problem analytically.

Following the work from Ref. [2], we introduce a rotated spin basis defined by

$$\tau_i^\pm = e^{\mp i\theta\sigma_y}\sigma_i e^{\pm i\theta\sigma_y},$$

with the rotation angle determined by $\tan 2\theta = \mp \frac{2g\alpha_s}{\Delta}$. This transformation diagonalizes the static spin-field interaction such that $\Delta\sigma_z \pm 2g\alpha_s\sigma_x = \tilde{\Delta}\tau_z^\pm$, where $\tilde{\Delta} = \sqrt{\Delta^2 + (2g\alpha_s)^2}$.

The Hamiltonian can be rewritten in this basis as:

$$H'_\pm = \omega a^\dagger a \pm \omega \alpha_s (a + a^\dagger) + g(a + a^\dagger) [\cos(2\theta) \tau_x^\pm \mp \sin(2\theta) \tau_z^\pm] + \tilde{\Delta} \tau_z^\pm, \quad (89)$$

where we have omitted the constant term $\omega \alpha_s^2$. We identify the unperturbed and interaction terms for a Schrieffer–Wolff (SW) transformation as $H_0 = \omega a^\dagger a + \tilde{\Delta} \tau_z^\pm$ and $H_1^\pm = \pm \omega \alpha_s (a + a^\dagger) + g(a + a^\dagger) [\cos 2\theta \tau_x^\pm \mp \sin 2\theta \tau_z^\pm]$.

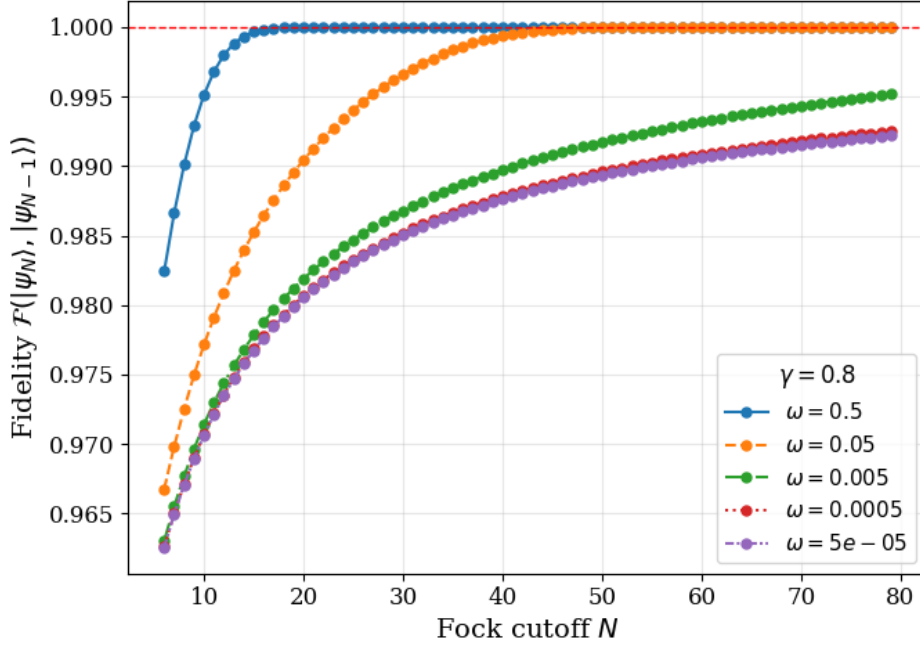


Figure 10: Fidelity between consecutive bosonic cutoffs for QRM displaced Hamiltonian written in squeezed basis. Δ is fixed to 1, as per usual.

To eliminate the odd contribution $H_1^\pm$ to first order, we seek an anti-Hermitian generator $s_1^\pm$ satisfying $[s_1^\pm, H_0] = -H_1^\pm$. We propose the ansatz,

$$s_1^\pm = aX - a^\dagger X^\dagger, \quad (90)$$

where X acts solely on the spin subspace. Solving the resulting operator equations yields the generator,

$$s_1^\pm = (a - a^\dagger) \left[\mp \alpha_s + \frac{g \sin(2\theta)}{\omega} \tau_z^\pm - \frac{\omega g \cos(2\theta)}{\omega^2 - 4\tilde{\Delta}^2} \tau_x^\pm \right] + i(a + a^\dagger) \frac{2\tilde{\Delta} g \cos(2\theta)}{\omega^2 - 4\tilde{\Delta}^2} \tau_y^\pm. \quad (91)$$

The first-order effective Hamiltonian is given by $H^{\text{SW}1} = H_0 + \frac{1}{2}[s_1, H_1]$. Calculating this commutator in the classical oscillator limit yields,

$$H_\pm^{\text{SW}1} = \omega a^\dagger a + \tilde{\Delta} \tau_z^\pm \mp \frac{4g^2 \alpha_s^2}{\tilde{\Delta}} \tau_z^\pm \pm \frac{g^2 \Delta^2}{\tilde{\Delta}^3} (a + a^\dagger)^2 \tau_z^\pm. \quad (92)$$

We aim to diagonalize the bosonic subspace via a Bogoliubov transformation, we will see why soon. Focusing on $H_+^{\text{SW}1}$,—the other option is equivalent— we analyze the stability of the projection into the eigenspaces of τ_z^+ . Projecting into the negative eigenspace

results in a coefficient for $a^\dagger a$ given by $A(\gamma) = \omega - 2\frac{g^2\Delta^2}{\tilde{\Delta}^3}$. This coefficient is negative in the superradiant phase, leading to unstable dynamics. Consequently, we must project into the positive eigenspace of τ_z^+ . Grouping terms to facilitate the Bogoliubov treatment, we obtain the effective Hamiltonian,

$$H_{\text{eff}} = \left(\omega + 2\frac{g^2\Delta^2}{\tilde{\Delta}^3} \right) a^\dagger a + \left(\frac{g^2\Delta^2}{\tilde{\Delta}^3} \right) (a^2 + (a^\dagger)^2). \quad (93)$$

This allows us to define the squeezing parameter $r = \frac{1}{4} \ln\left(1 + \frac{4g^2\Delta^2}{\omega\tilde{\Delta}^3}\right)$. Rewriting the displaced Hamiltonian in terms of the new bosonic operators b and $b^\dagger$ yields:

$$H_+ = \omega \cosh(2r) b^\dagger b - \frac{\omega}{2} \sinh(2r) (b^2 + (b^\dagger)^2) + e^{-r}(b + b^\dagger) [\omega\alpha_s + g(\cos(2\theta)\tau_x^+ - \sin(2\theta)\tau_z^+)] + \tilde{\Delta}\tau_z^+, \quad (94)$$

where constant shifts have been omitted. We see that this basis improves numerical fidelity convergence compared to simple displacement (see Fig 10). That change to a squeezed basis has the same motivation as the aforementioned displacement did; we are accommodating the basis we are using to the characteristics of the Hamiltonian.

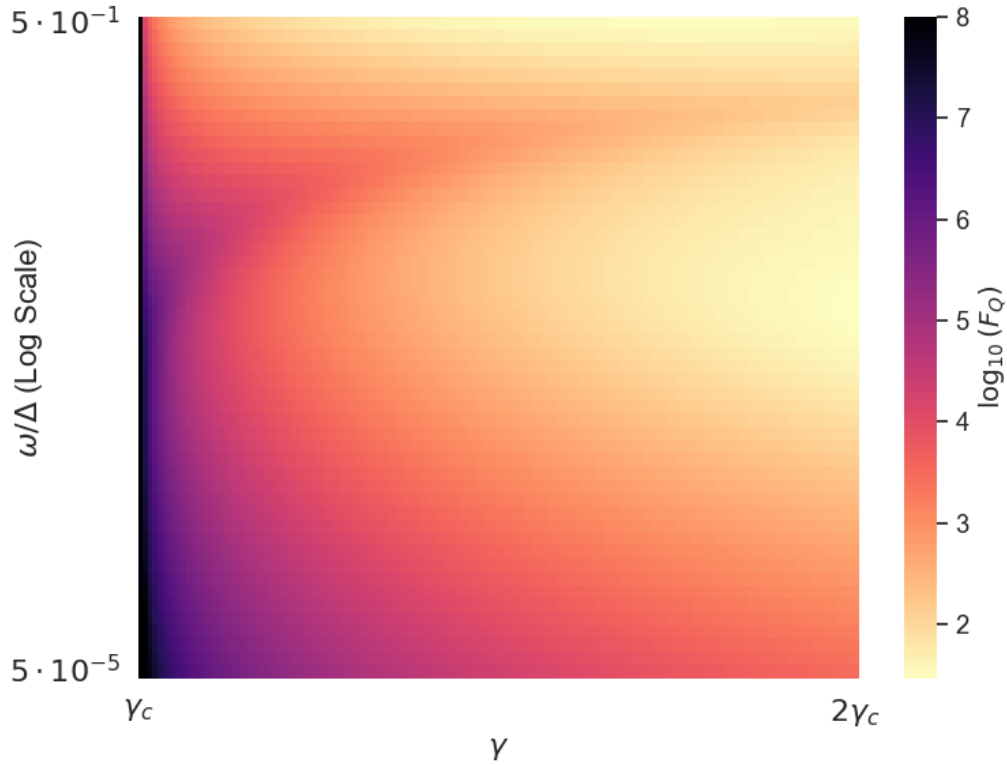


Figure 11: QFI heatmap in the superradiant phase. 50 ω/Δ bands represented in a log scale. x -axis in linear scale. Code in Ref. [6].

With this improved fidelity, we replicate the previous *modus-operandi* and compute the QFI for several ω/Δ bands, see Fig. 11. Here, we have intentionally clipped the QFI from above to preserve colour coherence in the graph. Regarding its content, we recover the divergence near the critical point as expected, but encounter a regime of

enhanced susceptibility in the top-right side of the graph that we have not been able to explain. Nevertheless, this regime is of great metrological interest. The explanation of this anomaly requires further investigation.

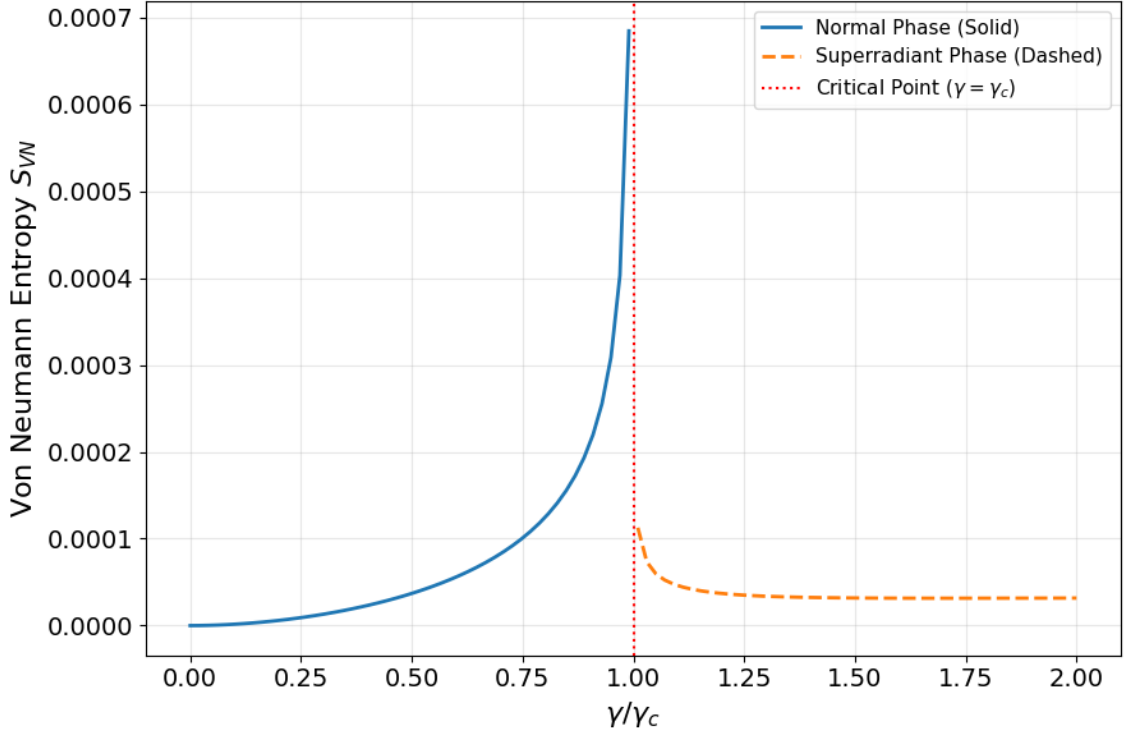


Figure 12: Entanglement entropy in both phases. $\omega/\Delta = 5 \cdot 10^{-5}$, $N = 50$. Code in Ref. [6].

7.3 Entanglement entropy

Having accurately defined the numerical GSs across both quantum phases, utilizing their respective optimal bases, we can evaluate other fundamental quantum information metrics. A key indicator of quantum criticality is the von Neumann entanglement entropy between the spin and the bosonic field, defined as $S_{vN} = -\text{Tr}(\rho_A \log_2 \rho_A)$, where ρ_A is the reduced density matrix of the atomic subsystem [30].

The entanglement entropy, computed independently in each phase using the respective optimized bases, displays a pronounced feature at γ_c (Figure 12). In the normal phase, S_{vN} grows monotonically and peaks sharply just below γ_c , while entering the superradiant phase is marked by a pronounced change in its slope, leading to a steep decline to a significantly lower value. This asymmetry provides an independent signature of the phase transition. Importantly, the overall magnitude of S_{vN} remains extremely small ($\lesssim 7 \times 10^{-4}$), confirming that the GS is well-approximated by a separable mean-field ansatz across both phases, consistent with the analytical predictions in the classical oscillator limit. The apparent discontinuity at γ_c itself should be interpreted with caution: since the normal- and superradiant-phase GSs are computed in different truncated bases (standard Fock vs. displaced-squeezed Fock), the gap must partly reflect a truncation artifact near the critical point rather than a true first-order character [31].

8 Conclusions

In this thesis, we have explored the landscape of critical quantum metrology, investigating how the diverging susceptibility associated with QPTs can be harnessed to surpass classical limits in parameter estimation. Specifically, we focused on the QRM, a finite-sized system that rigorously exhibits critical behavior in the classical oscillator limit $\omega/\Delta \rightarrow 0$. We reviewed the theoretical foundations of the QFI and the CRB as the ultimate metrics for estimation precision for such systems.

By analyzing the QRM, we successfully applied the SW transformation to derive low-energy effective Hamiltonians. The first-order correction (SW1) proved sufficient to identify the critical coupling strength $g_c = \sqrt{\omega\Delta/2}$ and analytically demonstrate the divergence of the QFI, scaling with a critical exponent of -2 . We extended this analysis via a second-order transformation (SW2), demonstrating that in the strict classical oscillator limit, the results formally converge with SW1. This indicates that higher-order corrections, while refining the model in the strong-coupling regime, do not alter the fundamental qualitative nature of the phase transition.

While these analytical methods proved robust in the normal phase, we identified a fundamental breakdown of the standard SW1 approach in the superradiant phase ($g > g_c$) due to the inversion of the effective potential. To overcome this limitation, we implemented a sophisticated numerical framework. By applying unitary displacement and squeezing transformations, we shifted the computational basis to accommodate the macroscopic mean-field and quantum fluctuations of the superradiant state. This strategy successfully restored numerical convergence, allowing us to strictly confirm the criticality and the QFI scaling on both sides of the transition. Furthermore, leveraging these optimized numerical wavefunctions, we evaluated the bipartite entanglement entropy, identifying a tendency change that serves as an independent and complementary signature of the quantum phase transition.

Overall, we have demonstrated that the finite-size QRM serves as a powerful and mathematically tractable platform for critical metrology, thereby validating the paradigm that finite-component systems can effectively mimic thermodynamic-limit criticality for metrological advantage.

Despite these highly promising theoretical bounds, this study focused entirely on the static-precision limit. A comprehensive assessment of critical metrology must inevitably address critical slowing down. This implies that adiabatic ground-state preparation requires an asymptotically diverging amount of time. Consequently, future research must incorporate time-dependent dynamics to determine whether the exponential precision gain fundamentally justifies the temporal cost of state preparation.

9 Sustainable Development Goals

This work is deeply rooted in fundamental science, which often advances knowledge without an immediate connection to applied objectives. For that reason, it is challenging to associate it directly with a specific Sustainable Development Goal (SDG). The UN SDGs are largely oriented toward practical, measurable improvements in areas like poverty reduction, health, education, clean energy, and climate action. Basic research,

by contrast, may not always produce tangible results in the short term, but it plays an indispensable role in enabling future technological and societal breakthroughs.

Nevertheless, the ultimate purpose of this research is to meaningfully benefit humanity and future generations. By advancing our understanding of the underlying physical principles that govern natural phenomena, the work creates the foundation upon which applied sciences and engineering solutions can be built. For instance, discoveries in quantum physics or materials science often appear abstract at first, yet they eventually lead to transformative innovations in medicine, communication, and sustainable energy. In this sense, fundamental science supports multiple SDGs indirectly: fostering innovation (SDG 9), providing the groundwork for quality education (SDG 4), and enabling long-term climate solutions (SDG 13).

Thus, while not aligned with a single SDG, this research embodies sustainability by expanding human knowledge globally.

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A Squeezing operator, Bogoliubov transform and diagonalization

We define the squeezing operator for a single mode of the electromagnetic field as [25]

$$S_z = e^{\frac{z a^\dagger a^\dagger - z^* a a}{2}}, \quad (95)$$

where $z = r e^{i\theta}$ is a complex number. Note, also, that the operator is unitary. The reason why we introduce this operator is that it allows us to construct a new set of bosonic operators

$$\tilde{a} = S_z^\dagger a S_z = a \cosh r + a^\dagger e^{i\theta} \sinh r, \quad (96)$$

$$\tilde{a}^\dagger = S_z^\dagger a^\dagger S_z = a^\dagger \cosh r + a e^{-i\theta} \sinh r, \quad (97)$$

this is called a Bogoliubov transform. It is notably useful when it comes to diagonalizing Hamiltonians of the following kind

$$H = \xi a^\dagger a + \kappa a^\dagger a^\dagger + \kappa^* a a. \quad (98)$$

From the expressions above, we see that the first term of the Hamiltonian transforms as

$$S_z^\dagger a^\dagger a S_z = (S_z^\dagger a^\dagger S_z)(S_z^\dagger a S_z) = (a^\dagger \cosh r + a e^{-i\theta} \sinh r)(a \cosh r + a^\dagger e^{i\theta} \sinh r), \quad (99)$$

where we have made use of the fact that the squeezing operator is unitary. We apply the same idea to the entire Hamiltonian

$$\begin{aligned} S_z^\dagger H S_z &= \xi(a^\dagger \cosh r + e^{-i\theta} a \sinh r)(a \cosh r + e^{i\theta} a^\dagger \sinh r) \\ &\quad + \kappa(a^\dagger \cosh r + a e^{-i\theta} \sinh r)^2 \\ &\quad + \kappa^*(a \cosh r + e^{i\theta} a^\dagger \sinh r)^2. \end{aligned} \quad (100)$$

Here, it is convenient to write κ using exponential notation, that is $\kappa = |\kappa|e^{i\phi}$. Making the choice $\phi = \theta$ notably simplifies the expansion, leading to

$$\begin{aligned} S_z^\dagger H S_z &= (\xi \cosh 2r + |\kappa| \sinh 2r) a^\dagger a \\ &\quad + \left(\frac{\xi}{2} \sinh 2r + |\kappa| \cosh 2r \right) (e^{i\phi} a^\dagger a^\dagger + e^{-i\phi} a a) \\ &\quad + |\kappa| \sinh 2r + \xi \sinh^2 r. \end{aligned} \quad (101)$$

We recall that the idea is to eliminate the non oscillator like terms, that is anything containing $a^\dagger a^\dagger$ and $a a$. We accomplish that by eliminating the second line of Eq. (101). This means enforcing that

$$\tanh(2r) = -\frac{2|\kappa|}{\xi}. \quad (102)$$

From this expression, we solve for r as

$$r = \frac{1}{2} \arctan \left(\frac{2|\kappa|}{\xi} \right), \text{ or } r = -\frac{1}{4} \ln \left(\frac{\xi - 2|\kappa|}{\xi + 2|\kappa|} \right). \quad (103)$$

Finally, the Hamiltonian becomes

$$H' = \tilde{\xi} a^\dagger a + \frac{\tilde{\xi} - \xi}{2}, \quad (104)$$

where $\tilde{\xi} = \sqrt{\xi^2 - 4|\kappa|^2}$. It fully resembles a quantum harmonic oscillator whose GS is $|0\rangle$, with a minimum energy of $E_0 = \frac{\tilde{\xi} - \xi}{2}$. Consequently, the eigenvalue equation is $S_z^\dagger H S_z |0\rangle = E_0 |0\rangle$. Multiplying by S_z from the left, we obtain $H(S_z |0\rangle) = E_0 (S_z |0\rangle)$, which formally proves that the exact GS of the initial Hamiltonian is $|GS\rangle = S_z |0\rangle$.